Part 14 — Core concepts for weeks 1 to 10: the biology an engineer needs, compiled

You asked why I could not simply write the roughly 900 pages of textbook reading that weeks 1 to 10 assign. There are three reasons, and only one of them is a limit. Copyright is not the limit: facts are not copyrightable, and what follows is original prose that states the same facts. The real limits are that I cannot check descriptive biology from memory — the arithmetic in this module was computed and re-verified, but "which enzyme catalyses which step" cannot be computed, and the more of it I write, the more unverifiable claims you inherit with no way to tell them from the verified ones; that I have no access to protein structures or micrographs, so the figures are schematics; and that volume is not the same as understanding, so 900 pages from me would be a worse copy of a book that has been proofread for decades.

What I can do well, and what this part does, is compile **the core ideas** of those ten weeks, in the form an engineer can use: each concept stated plainly, each derivable number computed, and the connection to the rest of the module made explicit. It is about 40 pages and about 1.7 times the density of part 1 (2,100 words a week against 1,270), and it is meant to be read *beside* Lehninger, Alberts and Brock, not instead of them: use it to know what matters, and the books for the mechanisms, structures and figures it deliberately leaves out.

How to read it

The dagger. Every constant marked † is a standard textbook value that I recalled rather than computed — a pK_a, a redox potential, a molecular count per cell, a reaction stoichiometry in a specific organism. They are listed with where to verify them in the table at the end. Everything *without* a dagger was either derived from the code or computed from the daggered inputs, and the computation is shown. So when a result looks surprising, check the daggered inputs first.

The eight ideas. The previous reply named the core concept — product flux is set by whichever constraint binds first — and eight ideas under it. This part supplies the biology each of them rests on:

Core idea	Biology it rests on	Chapters here
1 Mass balance and the yield ceiling	What a cell is made of and what it costs to make	B1, B5, B7
2 Redox and ATP balance	Cofactors, the proton motive force, respiration	B2, B4, B6
3 Thermodynamics decides direction	Free energy, coupling, the ATP hierarchy	B4, B5
4 Distributed control; regulation fights you	Enzyme kinetics and allostery; regulatory networks	B3, B8, B9
5 The cell is resource-limited	Ribosomes, proteome allocation, growth laws	B1, B8, B9
6 Transport and scale	Membranes, oxygen demand, respiratory quotient	B4, B6, B10
7 Measurement and the DBTL loop	Counting, selection, mutation and its statistics	B10
8 Economics	The carbon cost of biomass; why yield has a ceiling	B7

Each chapter ends with a short **what to carry forward** table and three self-check questions with answers. A glossary of 65 terms follows chapter B10, and it is the fastest way to find out whether you know a word the tracker assumes.

What is not here. Mechanisms drawn atom by atom, three-dimensional structures, every step of every pathway with its enzyme and cofactor, and the experimental history. Those are what the books are for. Where a chapter points you to one, it says which.

Ch.	Week	Chapter
B1	1	The cell as a reactor: contents, the central dogma and the genetic code, in numbers
B2	2	Proteins and enzymes: structure, charge, stability and catalysis
B3	3	Enzyme kinetics: the derivations and the regimes
B4	4	Bioenergetics: free energy, ATP, the proton motive force and the ATP yield
B5	5	Glycolysis and the pentose phosphate pathway
B6	6	The TCA cycle, respiration and overflow
B7	7	Biosynthesis: what a gram of cells costs
B8	8	Molecular biology: replication, transcription, translation and their regulation
B9	9	System-level regulation and the growth laws

B10	10	Microbial genetics and microbiology: growth, mutation, gene transfer, sterility
		Glossary · Recalled constants to verify

Chapter B1 · tracker week 1

The cell as a reactor: contents, the central dogma and the genetic code, in numbers

Part 1 described the cell as a reactor that builds its own catalysts. This chapter puts numbers on that sentence — how many molecules, how fast, at what cost — because every later calculation in the module uses one of them.

What is in a cell

An *E. coli* cell at moderate growth weighs about 0.3 pg dry†, of which roughly 55% is protein, 20% RNA, 9% lipid and 3% DNA, with the remainder lipopolysaccharide, peptidoglycan, glycogen and small molecules†. Its genome is 4.64 Mb† encoding about 4,300 proteins†, and about 88% of the sequence is coding † — bacterial genomes are compact, with short intergenic regions that carry the promoters, ribosome binding sites and terminators of part 1 chapter 8.

From those inputs, what follows is computed. Protein per cell is 0.3 × 0.55 = 0.165 pg. At an average 36 kDa† that is **2.8 million protein molecules**, or **9.0×10^8 amino-acid residues** at 110 Da each. Those two numbers are the denominators of most of what follows.

Copy number and concentration are the same quantity in different units. One molecule in a 1 fL cell is $1/(6.022 \times 10^{23} \times 10^{-15}$ L) = **1.66 nM**. So a regulatory protein present at ten copies is 17 nM, a metabolite at 1 µM is 600 molecules, and a strongly expressed enzyme at 30,000 copies is 50 µM. This conversion is why the repressor of part 8 chapter 27, with K = 10 nM and a handful of copies per cell, is in a sensible regime — and why a cell has no concentration to speak of for anything it holds fewer than a few tens of copies of; stochastic effects dominate there.

The central dogma, as an information-flow budget

DNA is replicated to DNA, transcribed to RNA, and translated to protein; information does not flow back from protein to nucleic acid. The exceptions are real and bounded: reverse transcription, RNA-dependent RNA replication in viruses, and catalytic RNA. For the organisms in this module the dogma holds as stated, and what matters for an engineer is that each arrow has a rate and a cost.

Process	Rate	Consequence
DNA replication	~1,000 nt/s per fork†, two forks	4.64 Mb ÷ (2 × 1,000) = 2,320 s = **39 minutes**
Transcription	40–80 nt/s†	A 1 kb gene is transcribed in 12–25 s
Translation	10–20 aa/s†	A 330-aa protein takes 17–33 s to synthesise
mRNA lifetime	half-life of a few minutes†	Transcriptional control acts within minutes
Protein lifetime	mostly longer than a generation	Loss of a protein is mainly dilution by growth

The replication arithmetic is worth dwelling on, because it is a computed result that matches a measured one: 39 minutes is almost exactly the 40-minute replication period (the C period) that part 9 chapter 32 used to compute how gene dose varies with position on the chromosome. And it explains something that looks paradoxical, a doubling time of 20 minutes in a cell whose chromosome takes 40 minutes to copy: the cell starts new rounds of replication before the previous round has finished, so a fast-growing cell carries overlapping replication forks and more copies of origin-proximal genes than terminus-proximal ones.

The genetic code, and what it implies for engineering

Sixty-four codons encode 20 amino acids and a stop: 61 sense codons and 3 stops. The mapping is degenerate — nine amino acids have two codons, one (isoleucine) three, five have four, three (leucine, serine, arginine) six, and two (methionine, tryptophan) only one. I built the standard table in code and asked what a random single-base substitution does to a sense codon. Of the 549 possible substitutions:

Outcome	Fraction	Reading
Synonymous — same amino acid	**24.4%**	Roughly one in four point mutations is silent at the protein level
Missense — different amino acid	**71.4%**	The large majority change the protein
Nonsense — premature stop	**4.2%**	A few per cent truncate it

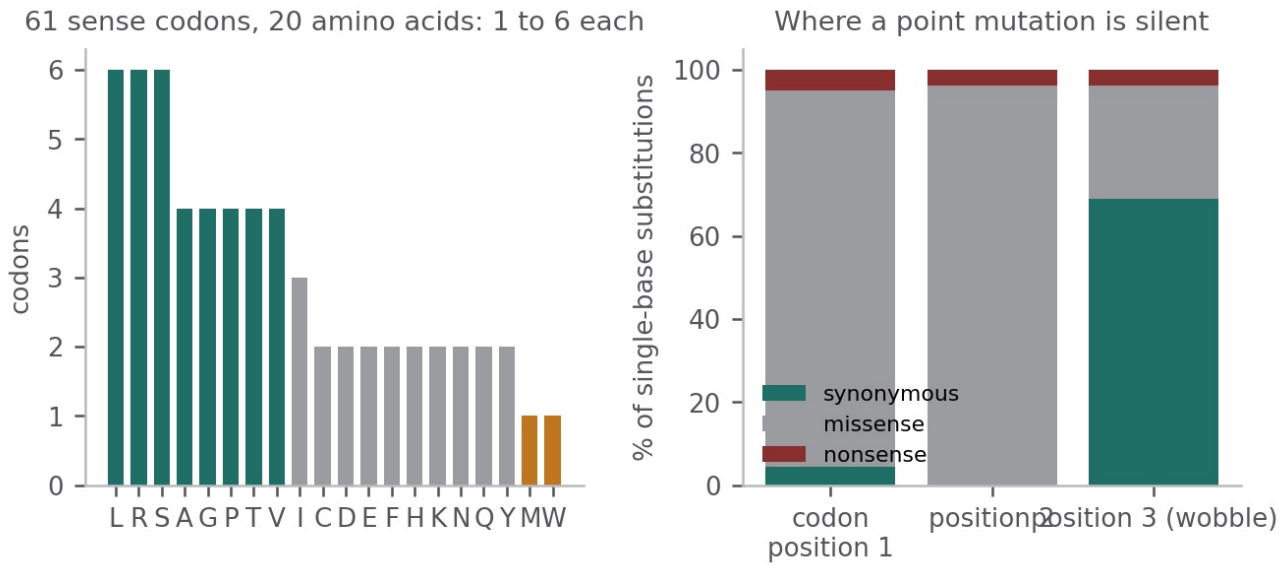


Figure B1.1 Left: codons per amino acid in the standard code, from 1 (Met, Trp) to 6 (Leu, Ser, Arg). Right: the fate of every possible single-base substitution by codon position — silent only at the third position, never at the second. Computed from the standard code for this module.

The position dependence is the useful part. **At the third position 68.9% of substitutions are synonymous; at the first, 4.4%; at the second, none.** That is the wobble: the third base is the least constrained, which is why codon usage and GC content are free to drift between organisms — a high-GC genome such as *Streptomyces* at 72% GC (part 8 chapter 31) is rich in G or C at the third position — and why a gene moved to a new host must be re-coded rather than just transferred. It also means a synonymous change, which is what domestication (part 7 chapter 21) and codon harmonisation (part 8 chapter 30) rely on, is almost entirely a third-position change.

Two caveats the textbook version skips. Synonymous does not mean neutral: codon choice affects translation speed, mRNA folding near the start codon, and in some genes co-translational folding, which is why part 8 chapter 30 treated "optimise the codons" as a hypothesis. And the code is not universal in every organism — mitochondria and some bacteria reassign codons — which is part of what makes genome recoding (part 7 chapter 25) possible at all.

What it costs to run: the ribosome budget

Here is the calculation that links this chapter to the growth laws of chapter B9. A cell must make all 9.0×10^8 residues of its protein once per doubling. At exponential growth rate μ, the required synthesis rate is μ times the protein content, so the number of ribosomes needed is

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N_ribosomes = μ · N_aa / (f_active · k_elongation)
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With f_{active} = 0.8 and k = 18 aa/s†, a cell growing at 0.6 h^{-1} needs about **10,500 ribosomes**. A cell growing at 2.0 h^{-1} is also bigger — taking twice the protein — and needs about **70,000**: a 3.3-fold faster growth rate costs a 6.7-fold larger ribosome population. Both agree with measured ribosome counts of order 10^4 and 7×10^4 at those rates †, which is a consistency check on the model rather than a measurement. It says the ribosome pool is superlinear in growth rate, so the cell commits a rapidly increasing share of its proteome to translation as it speeds up — and any pathway protein you add competes with it.

The energy bill is as follows. Each peptide bond costs about four high-energy phosphate bonds†: two to charge the tRNA (ATP to AMP is two equivalents, part 1 chapter 4), one GTP for EF-Tu delivery and one for EF-G translocation. A 330-aa protein therefore costs about **1,300 ATP-equivalents**, and the cell's protein polymerisation comes to 5 mmol of residues per gram dry weight × 4 = 20 mmol ATP-equivalents, against the 59.8 mmol the core model of part 10 charges as growth-associated maintenance. So **protein synthesis is about a third of the growth-associated energy budget** by this count — before the cost of making the amino acids, which chapter B7 adds.

Prokaryote and eukaryote, as far as an engineer needs

	Bacteria	Yeast and other eukaryotes
Genome organisation	One circular chromosome; genes grouped into operons, transcribed together as polycistronic mRNA	Several linear chromosomes; one gene per mRNA
Introns, splicing	Essentially none	Present; splicing before export

Transcription–translation	Coupled — ribosomes load while the mRNA is still being made	Separated by the nuclear envelope
mRNA processing	None; the 5′ end is the start	5′ cap, poly(A) tail
Replication origins	One	Many per chromosome
Compartments	Cytoplasm only (periplasm in Gram-negatives)	Nucleus, mitochondria, ER, Golgi, peroxisomes, vacuole

The coupling of transcription and translation is why a bacterial promoter and ribosome binding site are two independent dials on expression (part 1 chapter 8) and why a polar mutation in an operon can silence genes downstream. The compartmentalisation is why a pathway that works in *E. coli* may need re-engineering in yeast — acetyl-CoA is mitochondrial, as part 6 chapter 58 showed.

Cell contents	`0.3 pg dry weight; 55% protein → 2.8 × 10⁶ proteins, 9.0 × 10⁸ residues`
Copy number ↔ concentration	`1 molecule in 1 fL = 1.66 nM; 600 molecules = 1 µM`
Replication time	`genome ÷ (2 forks × 1,000 nt/s) = 39 min`
Point mutations, standard code	`24.4% synonymous, 71.4% missense, 4.2% nonsense; third position 68.9% silent`
Ribosomes required	`µ·N_aa/(f·k): ~10,500 at 0.6 h⁻¹, ~70,000 at 2.0 h⁻¹`
Cost of a protein	`~4 ATP-equivalents per residue: ~1,300 for 330 aa`

Common confusions

- **"Silent mutations do nothing."** They are silent in the amino-acid sequence only; they change translation speed and mRNA structure.
- **"Concentration is the natural unit."** At a few tens of copies per cell it is not; count molecules.
- **"A 20-minute doubling time means a 20-minute replication."** Replication takes 40 minutes; overlapping rounds are what let the cell double faster.

Self-check — answer on paper first

1. A repressor is present at 30 copies per cell. What is its concentration, and what does that imply about its operator binding?

30 × 1.66 nM = 50 nM. If the dissociation constant for its operator is at or below that, the repressor is largely bound; if it is far above, the operator is rarely occupied. That ratio, not the absolute count, sets the fold repression of part 8 chapter 27, and at 30 copies the cell-to-cell variation is substantial because a Poisson count of 30 has a standard deviation of 5.5 — about 18%.

2. Why do point mutations in a protein-coding gene so rarely give a synonymous change at the second codon position?

Because every change at the second base alters the amino acid in the standard code — the computed fraction is exactly zero. The second position largely determines the chemical class of the residue (U at that position gives the hydrophobic amino acids), so a change there almost always moves the residue into a different class, which is why the position is conserved more strongly than the third.

3. A colleague says a cell making a 36 kDa protein at 1% of its proteome needs only a trivial amount of energy. Check it.

1% of 2.8 × 10^6 proteins is 28,000 copies, each costing about 1,000 to 1,300 ATP-equivalents to synthesise, so about 3 × 10^7 ATP-equivalents per generation — about 1% of the protein-synthesis budget, and therefore about 0.3% of the growth-associated total. It is small in energy and not small in ribosome time, because the ribosomes that make it are not making growth proteins, which is why burden shows up as a growth penalty rather than an energy penalty.

What to open a book for. Alberts, *Molecular Biology of the Cell*, chapters 1 to 7 for the cell and the dogma, read for the figures; BioNumbers for every daggered constant in this chapter; Neidhardt's *Escherichia coli and Salmonella* for the definitive composition tables. MIT OCW 7.05 lectures 1 and 2 cover this chapter free.

Proteins and enzymes: structure, charge, stability and catalysis

A protein is a chain of twenty kinds of building block that folds because it is cheaper in free energy to fold than not to — by a margin so small that a single mutation can cost a tenfold change in how much is unfolded.

The twenty amino acids, grouped by what they do

All twenty share a backbone — an amino group, a carboxyl group and an α-carbon — and differ in the side chain. The grouping that predicts behaviour is chemical, and part 1 chapter 2 gave it; here is the quantitative layer that decides what you can do with it.

Class	Residues	Side-chain pK_a†	Where it sits and what it does
Non-polar	G A V L I M P F W	—	Buried in the core; the hydrophobic effect drives folding; proline breaks helices, glycine allows tight turns
Polar, uncharged	S T N Q C Y	Cys 8.3; Tyr 10.1	Surface and active sites; Ser, Thr, Tyr, Cys are the usual nucleophiles and modification sites
Acidic	D E	3.9; 4.2	Negative at physiological pH; salt bridges, metal binding
Basic	K R H	Lys 10.5; Arg 12.5; His 6.0	Positive at physiological pH; His is the one residue near neutral and is therefore a catalytic acid–base

The backbone terminals ionise too, near pK_a 2 for the carboxyl and 9 for the amino group†, though inside a chain those groups are consumed in peptide bonds and only the two ends remain.

Charge is a titration curve, and the arithmetic is useful

A group with pK_a p is protonated to the fraction $1/(1 + 10^{pH - p})$, so the net charge of a protein at a given pH is a sum of such terms. Writing that sum for a few sequences and finding where it crosses zero gives the isoelectric point, pI:

Sequence	pI	Net charge at pH 5	at 7.5	at 8
20 residues, one of each	7.11	+1.12	−0.14	−0.43
DDEEDEGSDE (acidic)	2.51	−7.16	−8.03	−8.09
KRKRHKRGSK (basic)	12.81	+7.91	+7.00	+6.91
HHHHHH (a His_6 tag)	7.90	+5.46	+0.15	−0.03

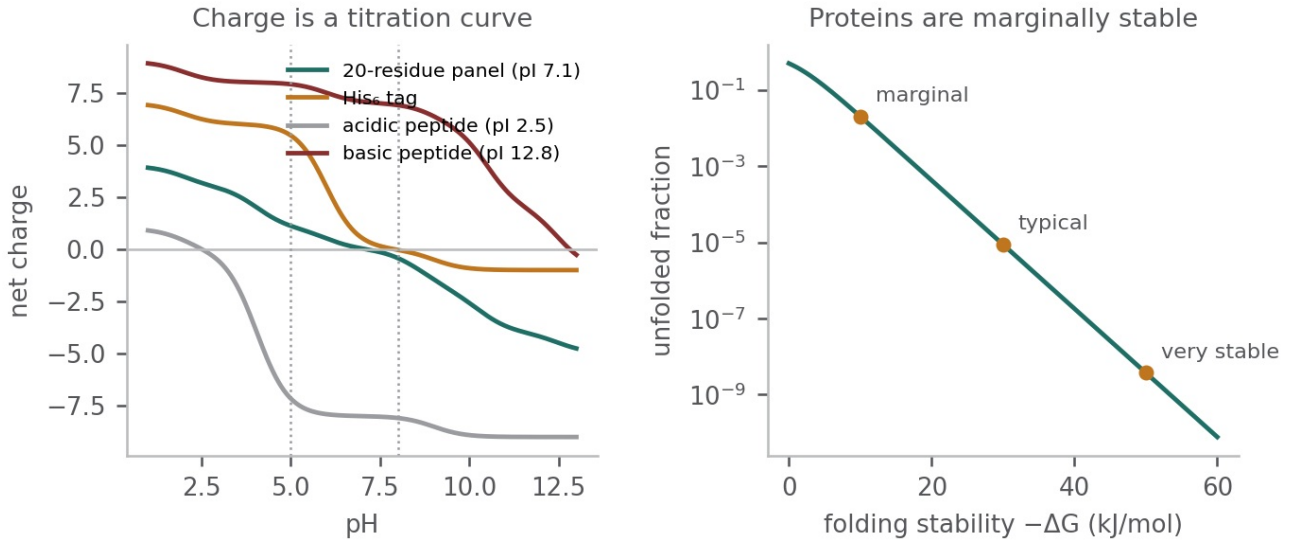


Figure B2.1 Left: net charge against pH for four peptides, with pH 5 and pH 8 marked. Right: fraction of a protein unfolded against its folding stability, with three typical stabilities marked. Computed from the pK_a values in the table and from $K = \exp(-\Delta G/RT)$ at 310 K.

The His_6 row is the one that earns its keep. A histidine is neutral — and therefore able to coordinate a nickel ion on a

resin — when it is deprotonated: at pH 8 that is 99% of residues, at pH 5 only 9%. So a His-tagged protein binds a nickel resin at pH 8 and is eluted by dropping the pH toward 5 (or by competing imidazole), which is the arithmetic behind a purification step that every protein lab runs without computing it. The same table says why a protein at its pI aggregates — net charge zero removes electrostatic repulsion — and why a buffer pH within a unit of the pI is a poor choice for storage.

Structure, in the four levels that matter

Primary structure is the sequence. **Secondary** structure is the local backbone pattern, stabilised by hydrogen bonds between backbone groups: the α-helix has 3.6 residues per turn, a rise of 1.5 Å per residue and a pitch of 5.4 Å, with each carbonyl bonded to the amide four residues further on †; the β-sheet has strands about 3.3 Å per residue joined side by side. A 20-residue helix is therefore 30 Å long, which is the thickness of a membrane's hydrophobic core — the reason transmembrane segments are about twenty residues of hydrophobic helix. **Tertiary** structure is the three-dimensional fold of one chain, driven by burial of hydrophobic side chains and held by hydrogen bonds, salt bridges and sometimes disulfides. **Quaternary** structure is the assembly of several chains, which is what makes cooperativity possible (chapter S3).

The peptide bond is planar, because the C–N bond has partial double-bond character, so the backbone's only freedom is rotation about two angles per residue, φ and ψ. The allowed combinations — the Ramachandran plot — are restricted by steric clashes, which is why helices and sheets exist at all and why proline, whose ring fixes φ, and glycine, which has no side chain to clash, behave differently.

Why proteins fold, and how marginal the result is

The dominant driving force is the **hydrophobic effect**: burying non-polar groups releases ordered water, and the entropy gain of that water pays for the loss of the chain's own conformational entropy. Against it, a folded protein's net stability is small — typically 20 to 60 kJ/mol† — the difference between two enormous numbers. The equilibrium constant is K = exp(−ΔG/RT), with RT = 2.58 kJ/mol at 310 K:

ΔG_{fold}	K = [folded]/[unfolded]	Unfolded fraction
−10 kJ/mol	48	2.0%
−20 kJ/mol	2,340	4.3×10^{-4}
−30 kJ/mol	114,000	8.8×10^{-6}
−50 kJ/mol	2.7×10^{8}	3.8×10^{-9}

Two consequences. A 5 kJ/mol destabilising mutation — one lost hydrogen bond, one hydrophobic residue exposed — **multiplies the unfolded fraction by exp(5/2.58) = 7.0**. A protein at −10 kJ/mol that loses 5 is at 13% unfolded, and an unfolded protein is a substrate for proteases and a seed for aggregation. Hence the engineering fact that most random mutations are destabilising, that a protein carries only a limited stability budget to spend on new function, and that stabilising mutations are valuable because they buy room for activity-improving ones.

The second is Levinthal's argument. A 100-residue chain with three conformations per residue has $3^{100} = 5 \times 10^{47}$ states, and sampling them at 10^{13} per second would take 5×10^{34} seconds, 10^{27} years. Real proteins fold in milliseconds to seconds, so folding cannot be a random search: the energy landscape is a funnel, and each local contact formed lowers the energy and narrows the search. Chaperones (GroEL/GroES, DnaK) do not direct folding; they prevent aggregation of intermediates and give a misfolded chain another attempt, which is why co-expressing them rescues some heterologous proteins (part 8 chapter 30).

Catalysis: how an enzyme lowers a barrier

Transition-state theory gives the rate of a reaction as $k = (k_B T/h)\cdot\exp(-\Delta G^{\ddagger}/RT)$. The prefactor $k_B T/h$ is 6.5×10^{12} s^{-1} at 310 K and is the maximum frequency at which any barrier can be crossed. Inverting it turns a rate into a barrier:

Rate constant	Barrier $\Delta G^{\ddagger}$
10^{-9} s^{-1} (an uncatalysed reaction taking decades)	129 kJ/mol
10^{-3} s^{-1}	94 kJ/mol
10^{3} s^{-1} (a good enzyme)	58 kJ/mol
10^{5} s^{-1}	46 kJ/mol

Each tenfold change in rate is RT ln 10 = **5.93 kJ/mol** of barrier, so a 10^{12} rate enhancement — a typical enzyme — is 71

kJ/mol, less than the energy of a few hydrogen bonds. That is the scale to hold in mind: catalysis is not a large energy effect, it is a small energy effect that is exponentiated. It is also why part 2 chapter 15's point that an enzyme lowers the barrier in both directions equally follows directly — the equilibrium constant depends on the difference between the reactant and product states, which the enzyme does not change.

The mechanisms, which the books treat atom by atom, fall into five groups: **proximity and orientation** — holding the reactants at the right geometry, so the entropic cost is paid at binding; **transition-state stabilisation** — the active site is complementary to the transition state, not the substrate; **general acid–base catalysis** — residues such as His, Asp and Glu move protons; **covalent catalysis** — a nucleophile such as Ser or Cys forms a transient bond; and **metal-ion catalysis** — stabilising charge, as magnesium does for every phosphoryl transfer.

Cofactors, and the vitamins behind them

Cofactor	Vitamin precursor†	What it carries
NAD^+/NADH, $NADP^+$/NADPH	Niacin	Hydride (two electrons); catabolic and anabolic respectively
FAD, FMN	Riboflavin	One or two electrons
Coenzyme A	Pantothenate	Acyl groups
Thiamine pyrophosphate	Thiamine	Aldehyde groups; the two-carbon unit in decarboxylations and transketolase
Pyridoxal phosphate	Vitamin B_6	Amino groups in transamination
Biotin	Biotin	CO_2 in carboxylations
Tetrahydrofolate	Folate	One-carbon units
Cobalamin	Vitamin B_{12}	Radical rearrangements and methyl groups
S-adenosylmethionine	(made from methionine and ATP)	Methyl groups
Haem; iron–sulfur clusters	—	Electron transfer; O_2 chemistry

Enzymes are named by the Enzyme Commission into seven classes†: oxidoreductases, transferases, hydrolases, lyases, isomerases, ligases, and, since 2018, translocases. An EC number such as 1.1.1.42 reads as class 1 (oxidoreductase), subclass 1 (acting on a CH–OH group), and so on down to the specific enzyme; it is the key to look an enzyme up in BRENDA.

Charge	`fraction protonated = 1/(1 + 10^(pH − pK_a)); sum over groups; pI where net = 0`
Helix geometry	`3.6 residues/turn, 1.5 Å/residue; 20 residues = 30 Å`
Folding	`K = exp(−ΔG/RT); −30 kJ/mol → 9 × 10⁻⁶ unfolded; a 5 kJ/mol loss → ×7.0`
Transition-state theory	`k = (k_BT/h)exp(−ΔG‡/RT); 5.93 kJ/mol per decade at 310 K`
A typical enzyme	`10¹² enhancement ≈ 71 kJ/mol of barrier`

Common confusions

- **"A stable protein is far from unfolding."** It is typically 20 to 60 kJ/mol from it, which is a few hydrogen bonds.
- **"The pI is where the protein is most stable."** It is where it is least soluble.
- **"Enzymes make reactions thermodynamically favourable."** They change the barrier and not the equilibrium.

Self-check — answer on paper first

1. A His_6-tagged protein will not bind the resin at pH 6.0. Why, and what do you change?

At pH 6 half of each histidine is protonated (the pK_a is 6.0), so only half of the imidazole groups are available to coordinate nickel and the tag's effective binding is far weaker than at pH 8, where 99% are neutral. Raise the pH to 7.5–8, and keep imidazole out of the binding buffer beyond a few millimolar. If the protein itself has a surface cluster of histidines or a low pI that makes it bind nonspecifically, the fix is the other way, in the wash.

2. A variant is 8 kJ/mol less stable than wild type and the wild type is −25 kJ/mol. What happens to the unfolded fraction?

Wild type: $K = \exp(25/2.58) = 1.6 \times 10^4$, unfolded fraction 6×10^{-5}. Variant at −17 kJ/mol: $K = \exp(17/2.58) = 720$, unfolded fraction 1.4×10^{-3}. The unfolded fraction rises 22-fold, from 0.006% to 0.14%. That is still small, which is why the variant may express and assay fine at 30 °C, and why it can fail in a fermenter at 37 °C with proteases present or on a modest rise in temperature, since the stability falls as the temperature approaches the melting point.

3. An enzyme speeds a reaction by 10^9. Convert to a barrier change and say what it implies about the binding energy.

9 decades × 5.93 kJ/mol = 53 kJ/mol. The enzyme must therefore bind the transition state more tightly than the substrate by about that much, which is the quantitative content of "the active site is complementary to the transition state". It also says a substrate-analogue inhibitor designed to resemble the transition state could in principle bind up to 10^9 times more tightly than the substrate does, which is how transition-state analogues achieve their very low dissociation constants.

What to open a book for. Lehninger chapters 3 to 6 for amino acids, structure and enzymes, and Alberts chapter 3 — both are worth reading for their structure figures, which nothing in this module replaces. Open one enzyme of your own pathway in Mol* on the PDB site and rotate it until the active site makes sense, as part 1's appendix suggested; it is the single best use of this week's reading time.

Enzyme kinetics: the derivations and the regimes

Michaelis–Menten is three lines of algebra and four pages of consequences. The algebra is in part 1 chapter 3; this chapter supplies the consequences — what the constants mean, when an assay is measuring them, and what changes with pH and temperature.

The derivation, and the assumption that matters

For $E + S \rightleftharpoons ES \rightarrow E + P$ with rate constants k_1, k_{-1} and k_{cat}, the Briggs–Haldane treatment sets $d[ES]/dt = 0$ and imposes the enzyme balance $E_{total} = [E] + [ES]$. The result is

$$v = V_{max}[S]/(K_M + [S]), \quad V_{max} = k_{cat} \cdot E_{total}, \quad K_M = (k_{-1} + k_{cat})/k_1$$

Michaelis and Menten's original derivation assumed instead that the binding step is at equilibrium, which gives $K_M = k_{-1}/k_1 = K_d$. The two agree only when $k_{cat} \ll k_{-1}$ — when catalysis is slow compared with dissociation. In general **K_M is not a dissociation constant**, and reading it as an affinity is wrong whenever k_{cat} is comparable with k_{-1}, which for fast enzymes it is. This is the first of part 1 chapter 3's "things people get wrong", and the derivation is why.

The three parameters answer three different questions. **k_{cat}**, the turnover number, is the maximum rate per active site — the capacity. **K_M** is the substrate concentration at half-maximal rate — where on the curve the cell is operating. And **k_{cat}/K_M** is the rate constant at low substrate, where $v = (k_{cat}/K_M)[E][S]$, so it is the figure of merit for an enzyme working below saturation, and it has an upper bound near 10^8 to 10^9 $M^{-1}s^{-1}$ set by how fast substrate and enzyme can meet by diffusion†. For scale, typical enzymes have a median k_{cat} around 10 s^{-1} and a median K_M around 100 µM, and so a median k_{cat}/K_M of order 10^5 $M^{-1}s^{-1}$† — three to four orders below the diffusion limit, which is the room protein engineering has to work in.

The regimes, and why the form of the rate law changes

Three regimes follow from the same equation. For $[S] \ll K_M$, $v \approx (V_{max}/K_M)[S]$ — first order in substrate, and linear in enzyme, so doubling the enzyme or the substrate doubles the rate. For $[S] \gg K_M$, $v \approx V_{max}$ — zero order in substrate, so the rate is set by enzyme amount alone and adding substrate does nothing. Near K_M it is mixed. The cell's intracellular concentrations usually sit around or below K_M, which is why part 2 chapter 15 found that substrate supply matters as much as enzyme amount.

The regime also decides how a reaction proceeds in time. Integrating the Michaelis–Menten equation has a closed form involving the Lambert W function, and evaluating it for V_{max} = 1 mM/min and K_M = 1 mM gives the time to convert 10% and 90% of the substrate:

Starting substrate	10% converted	90% converted	First-order estimate	Zero-order estimate
0.1 K_M	0.12 min	2.39 min	2.30 min	0.09 min
1 K_M	0.21 min	3.20 min	2.30 min	0.90 min
10 K_M	1.11 min	11.30 min	2.30 min	9.00 min

Read the columns together. At low substrate the reaction is first order and the 90% time is the first-order estimate, $\ln 10 \cdot K_M/V_{max}$; at high substrate it is close to the zero-order estimate, $0.9[S]_0/V_{max}$; in between neither is right. And the time to reach 10% conversion — the window within which an *initial rate* is valid — varies from 0.12 to 1.1 minutes across the three cases. That is why the standard instruction is to measure initial rates at less than 5–10% conversion and to check linearity of product against time, not to assume it: the window depends on $[S]_0/K_M$, and an assay designed at one substrate concentration is not valid at another.

Inhibition, and what the patterns mean

Type	Binds	Apparent K_M	Apparent V_{max}	Double-reciprocal lines
Competitive	Free enzyme, at the substrate site	Rises: $K_M(1 + [I]/K_i)$	Unchanged	Intersect on the 1/v axis
Uncompetitive	The ES complex only	Falls	Falls	Parallel
Non-competitive	E and ES equally, elsewhere	Unchanged	Falls	Intersect on the 1/[S] axis

Mixed	E and ES with different affinities	Changes	Falls	Intersect off both axes

The practical content is the one part 1 chapter 3 stated: the pattern tells you where the inhibitor acts, and competitive inhibition is overcome by substrate. A further kind needs one line because it appears in production hosts — **substrate inhibition**, $v = V_{max}[S]/(K_M + [S] + [S]^2/K_{si})$, where a second substrate molecule binds the ES complex unproductively. The rate peaks at $[S]^* = \sqrt{(K_M K_{si})}$, and at that point the enzyme reaches only $V_{max}/(1 + 2\sqrt{(K_M/K_{si})})$: for $K_M = 0.1$ and $K_{si} = 10$, the optimum is $[S] = 1$ and the maximum rate is 83% of V_{max}. A fed-batch that overshoots the substrate concentration lowers the rate, which is one more reason feed control matters.

Reversible reactions and the Haldane relation

An enzyme catalyses both directions, and its kinetic parameters in the two directions are not independent. The Haldane relation fixes the equilibrium constant in terms of them: $K_{eq} = (V_f K_{M,P})/(V_r K_{M,S})$. Its use is exactly the one part 2 chapter 16 gave: **take K_{eq} from thermodynamics and let the kinetics respect it**, rather than fitting a K_{eq} that the enzyme cannot violate. The net rate of a reversible enzyme is its forward rate times $(1 - Q/K_{eq})$, which goes to zero as the mass-action ratio reaches equilibrium — the kinetic expression of the point that a near-equilibrium step carries little net flux and cannot be a control point (part 2 chapter 14).

Temperature and pH

A rate constant follows the Arrhenius form $k = A \cdot \exp(-E_a/RT)$. The practical measure is Q_{10}, the factor per ten degrees, which for activation energy E_a between 30 °C and 40 °C is $\exp[E_a/R \cdot 10/(T_1 T_2)]$:

E_a	Q_{10} (30 to 40 °C)
30 kJ/mol	1.46
50 kJ/mol	1.88
80 kJ/mol	2.76

The commonly quoted "doubling per ten degrees" corresponds to about 50 kJ/mol and is a convenient average rather than a rule. The activity of a real enzyme then falls at higher temperature because the protein unfolds — chapter B2's marginal stability — so an activity–temperature curve is the product of a rising Arrhenius term and a falling folded fraction, with an optimum below the melting point. Activity against pH is a bell curve for a similar reason: the catalytic residues have pK_a values, the rate needs one protonated and one deprotonated, and the product of the two ionisation fractions peaks between the two pK_a values. Both curves are measured, not assumed, and the pH and temperature of an assay should be stated beside every K_M and k_{cat} you report.

What an assay actually measures

A kinetic constant is only meaningful with its conditions, and the list of what to fix is short: temperature, pH and buffer, ionic strength, the concentration of every cofactor and metal, enzyme concentration and purity, the substrate range — which must bracket K_M on both sides — and the conversion at which the rate was read. For a two-substrate enzyme the second substrate's concentration is part of the K_M (chapter S3). And in vitro constants differ from in vivo: the cytoplasm is crowded at 200 to 320 g/L of protein (part 1 chapter 1), so a K_M measured in dilute buffer is the starting point for a model, not an input to trust.

Michaelis–Menten	$v = V[S]/(K_M + [S])$; $K_M = (k_{-1} + k_{cat})/k_1 \neq K_d$ in general
Regimes	$[S] \ll K_M$: first order · $[S] \gg K_M$: zero order
Initial-rate window	10% conversion at 0.12 / 0.21 / 1.1 min for $S_0 = 0.1 / 1 / 10\ K_M$
Substrate inhibition	optimum $[S]^* = \sqrt{(K_M K_{si})}$; rate $V/(1 + 2\sqrt{(K_M/K_{si})})$
Haldane	$K_{eq} = V_f K_{M,P}/(V_r K_{M,S})$
Q_{10}	$\exp[E_a/R \times 10/(T_1 T_2)]$; 1.88 at 50 kJ/mol

Common confusions

- **"K_M is affinity."** It is $(k_{-1} + k_{cat})/k_1$; it equals K_d only when catalysis is slow.
- **"I measured the rate over five minutes."** If conversion exceeded 10%, you measured an average over a falling rate.
- **"The enzyme's optimum temperature is a constant."** It depends on the assay time, because unfolding is time-

dependent.

Self-check — answer on paper first

1. An enzyme has k_{cat} = 400 s^{-1} and K_M = 20 µM. Compute k_{cat}/K_M and compare it with the diffusion limit.

400 s^{-1} ÷ 2 × 10^{-5} M = 2 × 10^{7} $M^{-1}s^{-1}$. That is about 2% to 20% of the 10^{8} to 10^{9} diffusion limit, so it is already a very efficient enzyme: there is between about five- and fifty-fold of headroom left before the physical limit, and it is two orders of magnitude above the median 10^{5} — so a campaign to improve this enzyme further would be working near the physical limit, and the control-coefficient test of part 8 chapter 30 should come first.

2. You measure initial rates at [S] = 5 K_M over 3 minutes and see 40% conversion. What is wrong, and what do you change?

Initial-rate validity requires under about 10% conversion, and 40% conversion means most of the three minutes lay outside that window, so the reading is an average over a rate that has fallen substantially — the apparent rate is too low and the K_M derived from several such points will be biased. Shorten the assay or lower the enzyme concentration until conversion is under 10%, confirm product is linear in time, and only then fit the rates. Where the signal is too small at 10% conversion, use a more sensitive detection rather than a longer reaction.

3. A dehydrogenase shows a clear maximum in rate as the substrate is raised from 0.1 to 10 mM. Give two explanations and a test for each.

Substrate inhibition — a second substrate molecule binding the ES complex — which predicts an optimum at $\sqrt{(K_M K_{si})}$ and a smooth decline fitted by the substrate-inhibition equation; and a contaminant or ionic-strength effect from the high substrate concentration, which predicts the same decline when an unrelated salt or the substrate's counter-ion is added at the same level. Test the first by fitting both models and checking whether the substrate-inhibition term is significant; test the second by holding the substrate at its optimum and varying ionic strength with an inert salt.

What to open a book for. Cornish-Bowden, *Fundamentals of Enzyme Kinetics*, the early chapters on steady-state kinetics and inhibition — and, for the engineering view, Shuler, Kargi & DeLisa on enzyme kinetics, which connects directly to reactor design. Part 2 chapter 12 of this module gives the fitting code, and chapter S3 extends the story to two substrates.

Bioenergetics: free energy, ATP, the proton motive force and the ATP yield

The "ATP per glucose" in every textbook is an assumption about a membrane gradient, and the assumption can be computed. This chapter computes it, and in doing so shows that the 32 you were taught is the high end of a range that starts near 20.

Free energy, equilibrium, and the thing that decides direction

The standard free-energy change is related to the equilibrium constant by $\Delta G'^{\circ} = -RT \ln K_{eq}$, and the free-energy change at actual concentrations is $\Delta G' = \Delta G'^{\circ} + RT \ln Q$ (part 1 chapter 4). A reaction runs forward when $\Delta G'$ is negative, regardless of the sign of $\Delta G'^{\circ}$. Two implications to keep from the textbook: free energies are additive for coupled reactions, so an unfavourable step can be driven by a favourable one that shares an intermediate or a carrier; and a pathway's driving force is set by its worst step, which is the max–min idea of part 2 chapter 14.

The energy currency, ATP, is worth computing rather than quoting. With $\Delta G'^{\circ} = -30.5$ kJ/mol† for ATP hydrolysis, and illustrative cytoplasmic concentrations of 3 mM ATP, 0.5 mM ADP and 10 mM phosphate†:

```
ΔG′ = −30.5 + 2.58 · ln[(0.0005)(0.010)/(0.003)] = −30.5 − 16.5 = −47.0 kJ/mol
```

That is the −47 kJ/mol that part 1 chapter 4 quoted from a measurement in growing *E. coli*, and the computation shows why the real value exceeds the standard one: the cell keeps ATP far above equilibrium with ADP and phosphate, so the ratio term contributes another −16.5 kJ/mol. **The cell is not storing energy in ATP; it is maintaining a disequilibrium**, and the disequilibrium is what every ATP-driven reaction taps.

Reduction potentials, and what an electron is worth

For redox reactions the free energy is $\Delta G'^{\circ} = -nF\Delta E'^{\circ}$, with $F = 96.485$ kJ·V^{-1}·mol^{-1} and n the number of electrons. Standard potentials† arranged from most reducing to most oxidising set the direction of electron flow:

Couple	E'° (V)†	Role
NAD^+/NADH	−0.320	Catabolic electron carrier
$NADP^+$/NADPH	−0.324	Anabolic electron carrier — nearly the same potential, held at a very different ratio
Fumarate/succinate	+0.031	Anaerobic terminal acceptor; part of the TCA cycle
Ubiquinone/ubiquinol	+0.045	Membrane electron carrier
Cytochrome c (Fe^{3+}/Fe^{2+})	+0.254	Respiratory chain (in organisms that have it)
O_2/H_2O	+0.816	Terminal acceptor of aerobic respiration

NAD and NADP have almost identical standard potentials, which is the point of part 1 chapter 4's statement that they are "held at deliberately different ratios": since the chemistry cannot distinguish them, the cell does it with kinetics and with enzymes that discriminate the 2′-phosphate. For NADH oxidised by oxygen, $\Delta E'^{\circ} = 0.816 - (-0.320) = 1.136$ V and $\Delta G'^{\circ} = -2 \times 96.485 \times 1.136 =$ **−219 kJ/mol**: roughly four times the energy needed to make one ATP, so the respiratory chain breaks the drop into steps, each coupled to pumping protons.

The proton motive force

Respiration pumps protons out of the cell, building a gradient with two parts: an electrical potential across the membrane, Δψ, interior negative, and a chemical pH difference. The proton motive force is

```
Δp = Δψ − (2.303RT/F)·ΔpH      with ΔpH = pH_in − pH_out;  2.303RT/F = 61.5 mV per pH unit at 310 K
```

For a typical growing *E. coli* — Δψ = −140 mV, interior at pH 7.6 and exterior at 7.0†:

```
Δp = −140 − 61.5 × 0.6 = −177 mV      → energy per mole of protons returning = 96.485 × 0.177 = 17.1 kJ/mol
```

The two parts are in different proportions in different organisms, and acid-tolerant ones hold a large ΔpH with a small Δψ. The total, 177 mV, is what drives ATP synthesis, transport and the flagellar motor. That sharing is what part 1 chapter 4

meant by "when you export a product using a proton antiporter, you are spending the same gradient that would otherwise have made ATP".

The ATP synthase, and a limit you can compute

ATP synthase lets protons flow back through a rotating ring of c-subunits, and the number of protons per ATP is the ring's c-subunit count divided by three (three ATP per full rotation of the catalytic head): c_{10} → 3.33, c_8 → 2.67, c_{12} → 4.0†. Whether a ring is thermodynamically possible is a one-line check — the protons must deliver at least as much energy as the ATP needs:

Ring	H⁺/ATP	Energy delivered at Δp = 177 mV	ATP needs	Efficiency
c_8	2.67	46 kJ	47 kJ	**103% — impossible here**
c_{10} (*E. coli*)	3.33	57 kJ	47 kJ	83%
c_{12}	4.00	68 kJ	47 kJ	69%

The minimum is 47/17.1 = **2.75 protons per ATP** at this driving force, so a c_8 ring cannot make ATP at 177 mV. It works in mitochondria because their proton motive force and phosphorylation potential are both larger† — and the table is a clean demonstration that thermodynamics sets a floor on the stoichiometry, not just a direction on the reaction. It also shows the cost of a large ring: a c_{12} synthase is 69% efficient, spending more gradient per ATP, and trades efficiency for faster rotation or different regulation.

The ATP yield per glucose: an assumption with a range

Glucose is oxidised to six CO_2 with 4 ATP from substrate-level phosphorylation (2 in glycolysis, 2 in the TCA cycle), 10 NADH and 2 $FADH_2$ (chapters B5 and B6). The oxidative part depends on the P/O ratio — ATP made per pair of electrons — which is the protons pumped per electron pair divided by the protons per ATP. Taking $FADH_2$ at two thirds of NADH's yield because it enters the chain after the first proton pump:

```
ATP per glucose = 4 + 10·(P/O)_NADH + 2·(2/3)(P/O)_NADH
```

H⁺ pumped per NADH	H⁺ per ATP	P/O (NADH)	ATP per glucose
6	4.00	1.50	21.0
6	3.33	1.80	24.4
8	4.00	2.00	26.7
8	3.33	2.40	31.2
10	4.00	2.50	32.3
10	3.33	3.00	38.0

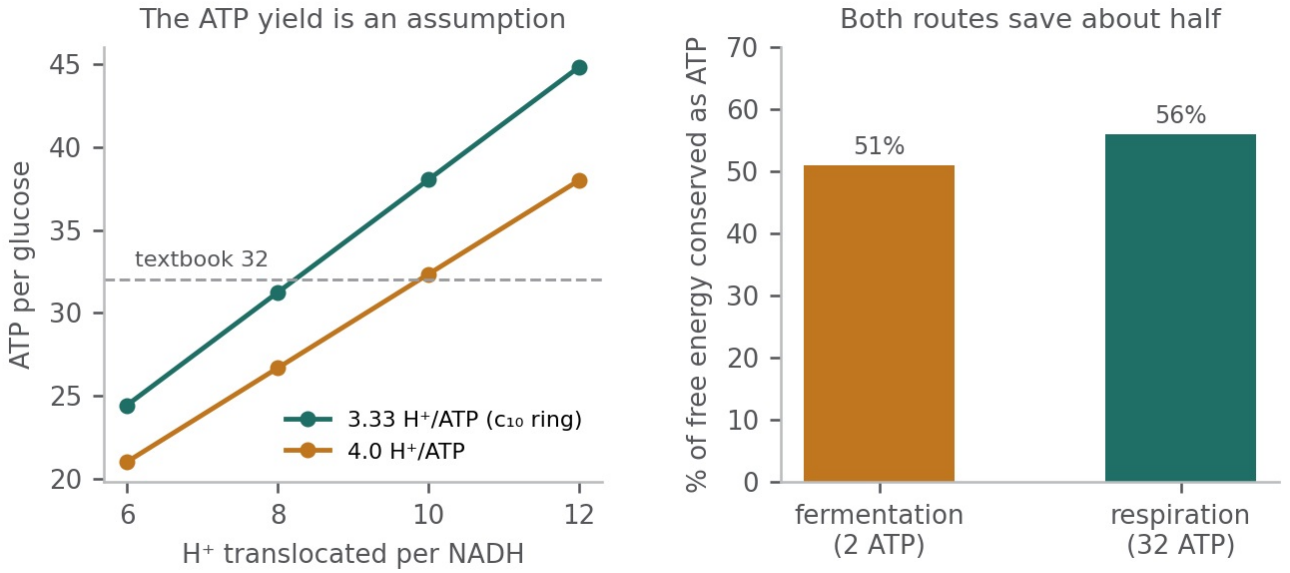


Figure B4.1 Left: ATP per glucose against the number of protons pumped per NADH, for two ATP synthase stoichiometries. Right: the share of the

glucose's free energy conserved as ATP by fermentation (2 ATP, ΔG ≈ −196 kJ/mol to lactate) and by respiration (32 ATP, ΔG ≈ −2,870 kJ/mol), at 50 kJ per ATP. Computed for this module.

The "32" is the row with P/O = 2.5 and 1.5, and it needs 10 protons per NADH and 4 per ATP. *E. coli* has a branched chain with variants that pump different numbers of protons — a proton-pumping NADH dehydrogenase and a non-pumping one, and terminal oxidases that differ† — so its effective yield depends on which are expressed, and that is a regulated choice. The consequence for the module is the one stated in part 1 chapter 4 and used in part 10: **treat the ATP yield as a stated modelling assumption and check how much your conclusion depends on it**. Between 21 and 32 ATP the oxidative branch's contribution changes by 50%, and a yield that scales with it moves with it.

One comparison puts the numbers in perspective. Fermentation to lactate conserves 2 × 50 = 100 kJ of the 196 kJ released, 51%; full respiration conserves 32 × 50 = 1,600 kJ of 2,870, 56%. **Both conserve about half the free energy**, which is a typical efficiency for a biological energy converter; respiration is not more efficient per joule, it is just able to extract far more of the substrate's energy because it removes the electrons rather than discarding them in a reduced end product.

Where the electrons go: NADH against NADPH, again

Because the carriers have the same potential but different cellular roles, converting between them costs something. Transhydrogenases move hydride between the pools; the membrane-bound one, PntAB, is driven by the proton motive force and pumps protons while making NADPH from NADH — which is the "energy cost" of part 2 chapter 19 stated in its mechanism — while the soluble one, UdhA, is the reverse direction and spends nothing. Expressing both strongly builds a futile cycle that dissipates the gradient, which is the trap that chapter named. Part 10's core model used an assumed 0.33 ATP per NADPH for PntAB†, which follows from roughly one proton per hydride at a few protons per ATP.

ATP in the cell	$\Delta G' = -30.5 + RT \ln([ADP][P_i]/[ATP]) = -47.0$ kJ/mol at 3, 0.5, 10 mM
Redox	$\Delta G^{\circ\prime} = -nF\Delta E^{\circ\prime}$; NADH + ½O₂: −219 kJ/mol
Proton motive force	$\Delta p = \Delta\psi - 61.5 \cdot \Delta pH$ (mV at 310 K) = −177 mV → 17.1 kJ/mol H^+
Minimum H^+/ATP	$\lvert\Delta G_{ATP}\rvert/(F\Delta p) = 2.75$; c_{10} ring = 3.33 (83% efficient)
ATP per glucose	4 + 10(P/O) + 1.33(P/O); 21 to 38 over the plausible range, 32 as the textbook case
Energy conserved	fermentation 51%, respiration 56% of the free energy released

Common confusions

- **"ATP stores energy."** It carries it, and the disequilibrium it maintains is the energy.
- **"A glucose yields 38 ATP."** That is the high edge of a range; 30 to 32 is the usual figure, and 21 to 26 is quite possible in a bacterium.
- **"NADH and NADPH are interchangeable because their potentials match."** The cell holds them at different ratios precisely because their chemistry cannot distinguish them.

Self-check — answer on paper first

1. A cell has Δψ = −110 mV and a ΔpH of 0 (neutral). What is the proton motive force, and can a c_{10} ATP synthase make ATP at ΔG = −47 kJ/mol?

Δp = −110 mV, so each mole of protons delivers 96.485 × 0.110 = 10.6 kJ. A c_{10} ring needs 3.33 protons per ATP, delivering 35.4 kJ against the 47 kJ the ATP requires — efficiency 133%, which is impossible, so the synthase would run backwards, hydrolysing ATP to pump protons. The minimum Δp that could drive it is 47/(3.33 × 96.485) = 146 mV, which is a useful number: it is why uncouplers, which dissipate the gradient, stop ATP synthesis even though the chain keeps running.

2. Your model assumes 32 ATP per glucose. Your organism pumps 6 H^+ per NADH and its synthase has a c_{12} ring. What do you correct, and by how much?

H^+ per ATP is 4.0, so $(P/O)_{NADH}$ = 6/4 = 1.5 and ATP per glucose = 4 + 10 × 1.5 + 1.33 × 1.5 = 21.0, a third lower than 32. Every quantity downstream that depends on ATP — the maintenance requirement in units of glucose, the growth yield, the oxygen demand — changes accordingly; a yield calculation that was ATP-limited falls by up to a third, and one that was carbon-limited barely moves, which is how to tell which regime you are in.

3. Why does fermentation conserve about the same fraction of free energy as respiration despite making 16 times less ATP?

Because it releases far less energy to begin with. Glucose to two lactate releases about 196 kJ/mol and conserves 100 kJ

of it, 51%; complete oxidation releases 2,870 kJ and conserves 1,600, 56%. The ATP number differs by a factor of 16 because the fraction of the substrate's energy that is released at all is 7%, the rest being left in lactate. The efficiency per joule is similar; the difference is how much of the substrate you are willing to leave unoxidised, which is the carbon-and-electron bookkeeping of part 2 chapter 18.

What to open a book for. Lehninger chapter 13 and the chapters on oxidative phosphorylation, read twice; Nicholls & Ferguson, *Bioenergetics*, for the proton motive force treated properly. Doran's energy-balance chapter does the accounting in engineering notation, and it is where problem-solving practice is.

Chapter B5 · tracker week 5

Glycolysis and the pentose phosphate pathway

Know the three irreversible steps, the net stoichiometry, and where the branch points sit. The rest of the pathway you can look up; those are what you will be asked to engineer around.

The pathway, by what each step does

Glycolysis converts one glucose (C_6) into two pyruvate (C_3) in ten steps, in two phases. The **preparatory phase** invests two ATP: glucose is phosphorylated (in *E. coli*, by the phosphotransferase system, which spends a phosphoenolpyruvate instead of an ATP), isomerised to fructose-6-phosphate, phosphorylated again by phosphofructokinase, and cleaved by aldolase into two triose phosphates that interconvert. The **payoff phase** oxidises glyceraldehyde-3-phosphate with NAD^+ and phosphate to 1,3-bisphosphoglycerate, then to 3-phosphoglycerate (making ATP), to 2-phosphoglycerate, to phosphoenolpyruvate, and finally to pyruvate (making ATP). The net equation is the one in part 1 chapter 5:

$$\text{glucose} + 2\ NAD^+ + 2\ ADP + 2\ P_i \rightarrow 2\ \text{pyruvate} + 2\ NADH + 2\ ATP + 2\ H^+ + 2\ H_2O$$

Step	Enzyme	Character
1	Glucose phosphorylation (PTS in *E. coli*; hexokinase in yeast)	**Irreversible** — commits glucose; regulated
2	Phosphoglucose isomerase	Near equilibrium
3	Phosphofructokinase	**Irreversible** — the main control point; allosterically regulated
4	Aldolase	Near equilibrium
5	Triosephosphate isomerase	Near equilibrium
6–7	GAPDH; phosphoglycerate kinase	Near equilibrium; make NADH and the first ATP
8–9	Phosphoglycerate mutase; enolase	Near equilibrium
10	Pyruvate kinase	**Irreversible** — regulated; makes the second ATP

The three irreversible steps are the ones that sit far from equilibrium (part 2 chapter 14), they are the points at which the pathway is regulated, and in gluconeogenesis they are bypassed by different enzymes — fructose-1,6-bisphosphatase, and phosphoenolpyruvate synthetase or carboxykinase — which is why a cell can run glycolysis forwards and backwards without a futile cycle, provided both sets are not on at once. The seven near-equilibrium steps are the ones that cannot be control points.

Alternatives that exist for a reason

The Entner–Doudoroff pathway converts glucose-6-phosphate to 6-phosphogluconate, dehydrates it to 2-keto-3-deoxy-6-phosphogluconate and cleaves that into pyruvate and glyceraldehyde-3-phosphate. Its net yield is **1 ATP, 1 NADH and 1 NADPH** per glucose† — half the ATP, but with NADPH generated for free — and, as part 1 chapter 5 reported, the pathway needs several-fold less enzyme protein for the same flux. That is the trade: ED is the better choice when protein is expensive and ATP is not, which is why it is common in organisms growing aerobically on rich carbon. Phosphoketolase pathways cleave a pentose or hexose directly to acetyl phosphate and a triose, avoiding the decarboxylation at pyruvate dehydrogenase — which is why they are engineered in when acetyl-CoA yield is the target.

The pentose phosphate pathway, as a bookkeeping exercise

The oxidative branch converts glucose-6-phosphate to ribulose-5-phosphate in two oxidations and one decarboxylation, **making 2 NADPH and releasing 1 CO_2**. The non-oxidative branch — transketolase and transaldolase — shuffles carbon between sugar phosphates of three to seven carbons and is reversible, so it can run in either direction depending on demand. Its net stoichiometry, when the cell needs only NADPH, is the cycle

$$6\ \text{glucose-6-P} + 12\ NADP^+ \rightarrow 5\ \text{glucose-6-P} + 6\ CO_2 + 12\ NADPH + P_i$$

which is the clean form of "every glucose that goes round loses all six carbons as CO_2 and yields twelve NADPH" — in practice the cycle recycles carbon back into glycolysis, so the carbon is not lost on the first pass, but the decarboxylation is. When the cell needs ribose-5-phosphate for nucleotides, the non-oxidative branch runs from fructose-6-phosphate and glyceraldehyde-3-phosphate backwards and the oxidative branch is bypassed; when it needs erythrose-4-phosphate for

aromatics, a transketolase makes it (part 1 chapter 7).

How much NADPH, and so how much carbon

The calculation worth carrying out here is the one that explains why the oxidative branch matters, using numbers computed in part 10. The core model's biomass reaction demands 13.03 mmol of NADPH per gram dry weight†. At the growth rate of 0.754 h^{-1} that model reaches, the demand is 13.03 × 0.754 = **9.8 mmol NADPH per gram per hour**. If the oxidative pentose phosphate pathway were the *only* source, at 2 NADPH per glucose it would have to carry 4.9 mmol of glucose per gram per hour — **49% of the 10 mmol uptake**. That is an upper bound, because real cells make NADPH elsewhere too: isocitrate dehydrogenase in the TCA cycle, NADP-dependent malic enzyme, and the transhydrogenase of chapter B4. The measured oxidative-branch share in *E. coli* is a fraction of the bound†, and the bound is what makes the cofactor accounting of part 2 chapter 19 a real constraint rather than a bookkeeping detail: a product needing four NADPH per molecule on top of the biomass demand competes for a supply that is already half-committed.

The carbon cost follows directly. Each NADPH from the oxidative branch costs half a carbon, so the biomass demand alone, 13.03 mmol NADPH per gram, would cost 6.5 mmol of carbon per gram if met entirely there. Against the roughly 40 mmol of carbon in a gram of biomass† (chapter B7), that is 16% — the size of the penalty that makes the pentose phosphate route "the NADPH tap" and why deleting phosphoglucose isomerase, which forces all glucose through it, cost a third of the growth rate in part 10's deletion scan.

Regulation, and what to engineer around

Phosphofructokinase in *E. coli* is activated by ADP and inhibited by phosphoenolpyruvate† — the second is feedback from the pathway's own product, which keeps the pool of PEP from collapsing. The phosphotransferase system couples glucose uptake to the PEP/pyruvate ratio and is the entry point of catabolite repression (chapter B9). Glucose-6-phosphate dehydrogenase, the first oxidative step, is inhibited by NADPH, so the cell's NADPH demand regulates its own supply: when NADPH is consumed the enzyme speeds up. That coupling is why a strain with a high NADPH demand pulls carbon through the oxidative branch without any genetic change, as part 1 chapter 5 noted.

Glycolysis, net	glucose + 2 NAD^+ + 2 ADP + 2 P_i → 2 pyruvate + 2 NADH + 2 ATP
Irreversible steps	glucose phosphorylation, phosphofructokinase, pyruvate kinase
Entner–Doudoroff	1 ATP, 1 NADH, 1 NADPH per glucose; several-fold less enzyme protein
Oxidative PPP	glucose-6-P → ribulose-5-P: 2 NADPH + 1 CO_2
NADPH demand	13.03 mmol/gDW × 0.754 h^{-1} = 9.8 mmol/gDW/h; oxidative PPP alone would need 49% of uptake
Carbon price of NADPH	0.5 C per NADPH; 6.5 mmol C/gDW if all from the oxidative branch

Common confusions

- **"Glycolysis makes a lot of ATP."** Two net per glucose; respiration makes most.
- **"The PPP is a nucleotide pathway."** Its oxidative branch is the main NADPH source, and the non-oxidative branch supplies pentoses and erythrose-4-phosphate.
- **"Every enzyme of glycolysis is a candidate for overexpression."** Seven of the ten are near equilibrium and carry little control.

Self-check — answer on paper first

1. Why does the PTS-based uptake in *E. coli* give the same net ATP as hexokinase, and why does it matter anyway?

Hexokinase spends one ATP phosphorylating glucose, giving net 2 ATP overall from the 4 made minus the 2 spent in the preparatory phase. The PTS spends one PEP instead; a PEP would otherwise have made an ATP at pyruvate kinase, so the opportunity cost is the same one ATP and the net is again 2. What differs is *which pool the cost comes out of*: PEP is also the precursor for aromatic amino acids and for the anaplerotic carboxylation, so PTS-based uptake competes directly with those, which is part 1 chapter 7's point about aromatic products impairing glucose uptake.

2. A product needs 4 NADPH per molecule. Estimate how much glucose passes through the oxidative branch per molecule if no other NADPH source were available, and say what that does to the carbon yield.

At 2 NADPH per glucose through the oxidative branch, four NADPH need two glucose. Each loses a carbon to CO_2 on the first pass, so 2 of the 12 carbons are decarboxylated before the remainder returns to glycolysis and is used. The yield ceiling falls from the naive carbon limit — part 1 chapter 5 gave 0.81 mol/mol for lysine instead of 1.0 — and the correct answer depends on how much NADPH can be sourced elsewhere, which is why a transhydrogenase or a cofactor swap is

attractive.

3. Why is an irreversible step a good place to look for control, and a near-equilibrium step not?

At a near-equilibrium step, forward and reverse fluxes are both large and nearly equal, and the net flux is their small difference, set by how far the mass-action ratio sits from K_{eq}; raising the enzyme raises both fluxes and leaves the net almost unchanged. An irreversible step has a negligible reverse flux, so the net flux is set by the enzyme's own activity and its regulation — which is also why that is where the cell puts its regulators, and where an intervention can change flux.

What to open a book for. Lehninger chapters 14 and 15: redraw the pathway from memory on a blank sheet afterwards, which is the one piece of memorisation in the curriculum that pays. For ED and the alternative pathways, Flamholz et al., PNAS 110:10039 (2013). Stephanopoulos chapters 2 and 3 for the stoichiometric accounting that part 2 chapter 18 applies.

The TCA cycle, respiration and overflow

The cycle has two jobs — oxidise acetyl-CoA to feed the chain, and supply precursors for biosynthesis — and they pull against each other. Overflow metabolism is what the cell does when the supply of carbon outruns the cell's capacity to respire it.

The cycle, as carbon bookkeeping

Pyruvate is oxidised to acetyl-CoA by pyruvate dehydrogenase, releasing one CO_2 and one NADH. Acetyl-CoA (C_2) condenses with oxaloacetate (C_4) to form citrate (C_6), which is isomerised to isocitrate and then oxidatively decarboxylated to α-ketoglutarate (C_5), losing a CO_2 and making an NADH. α-Ketoglutarate is oxidatively decarboxylated again to succinyl-CoA (C_4), losing a second CO_2 and making another NADH. The remaining steps convert succinyl-CoA to succinate (making one GTP or ATP by substrate-level phosphorylation), then to fumarate (making $FADH_2$, or reducing quinone), to malate, and back to oxaloacetate (making a third NADH).

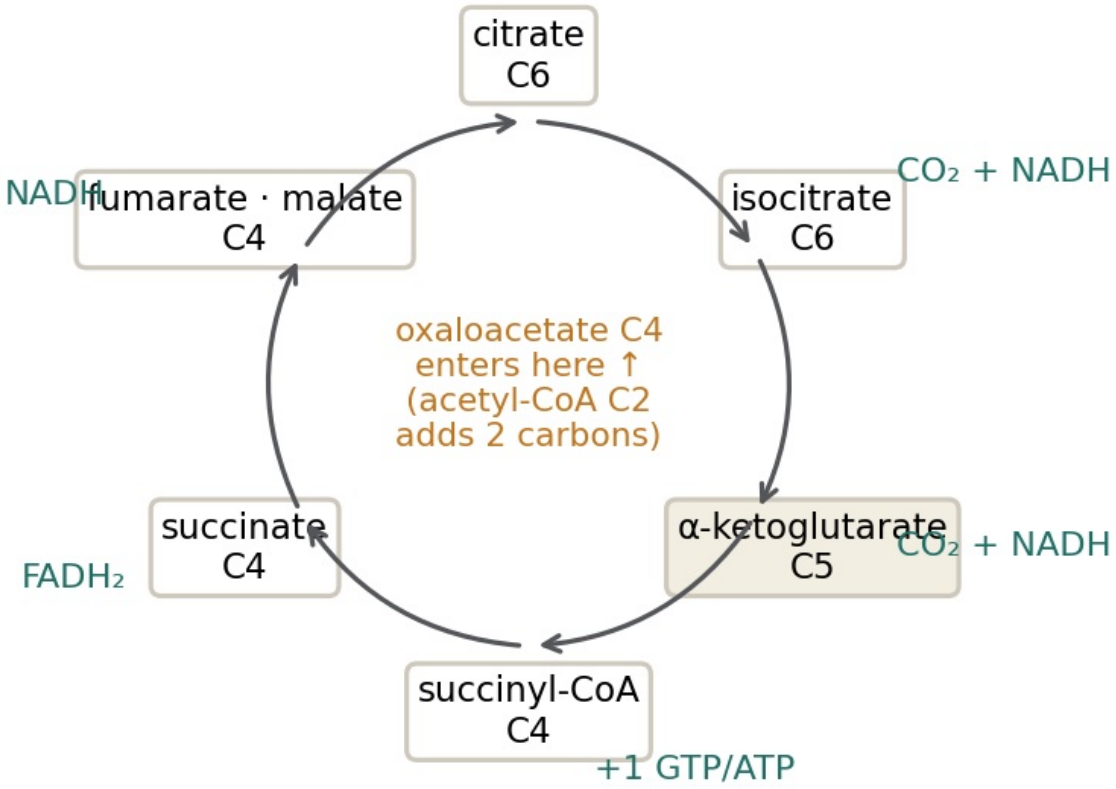


Figure B6.1 The TCA cycle as a carbon ledger: carbon counts at each intermediate, the two CO_2 released, and where reducing equivalents form. α-Ketoglutarate, shaded, is the nitrogen entry point. Drawn for this module.

Per turn of the cycle, the net is: acetyl-CoA + 3 NAD^+ + FAD + GDP + P_i + 2 H_2O → 2 CO_2 + 3 NADH + $FADH_2$ + GTP + CoA. Adding pyruvate dehydrogenase and glycolysis gives the totals that chapter B4 used: glucose → 6 CO_2 + **10 NADH + 2 $FADH_2$ + 4 ATP** (2 NADH and 2 ATP from glycolysis, 2 NADH from pyruvate dehydrogenase, 6 NADH, 2 $FADH_2$ and 2 ATP from two turns of the cycle).

A fact worth knowing because it surprises: the two carbons released as CO_2 in a turn are **not** the two that entered as acetyl-CoA in that turn. Citrate is a symmetric molecule, but aconitase treats its two arms differently, so it is the carbons that came from oxaloacetate that are lost; the acetyl-derived carbons are carried into succinate, which is symmetric and scrambles them, and are released only in later turns. It matters in practice for labelling experiments (part 11 chapter 43): a label on acetyl-CoA reaches CO_2 with a delay, not immediately.

The cycle as a supplier of precursors

Withdrawing an intermediate for biosynthesis drains the cycle: α-ketoglutarate to glutamate and the glutamate family; oxaloacetate to aspartate and the aspartate family; succinyl-CoA to lysine and methionine precursors and to tetrapyrrole synthesis; citrate to fatty acids in organisms that export it. To keep the cycle turning, **anaplerotic reactions** replenish oxaloacetate: PEP carboxylase (PEP + CO_2 → oxaloacetate, costing no ATP but consuming the PEP), pyruvate carboxylase in organisms that have it — *Corynebacterium* and *Bacillus*, though not *E. coli*† — which costs an ATP, and the glyoxylate shunt when growing on two-carbon substrates, which bypasses both decarboxylations (isocitrate lyase and malate synthase) and converts two acetyl-CoA into one four-carbon unit.

The engineering fact is part 1 chapter 6's: the cycle is a recycle loop with a purge, and the purge rate sets the required make-up. A product drawn from α-ketoglutarate or oxaloacetate competes with the cycle for carbon and with PEP for its anaplerotic route, and the yield is limited by the anaplerotic capacity and its ATP cost as much as by the pathway installed.

One note on a correction. Part 1 chapter 6 said succinyl-CoA feeds haem synthesis. That is true for yeast, animals and some bacteria, which make δ-aminolevulinate from succinyl-CoA and glycine (the Shemin route); *E. coli* and most bacteria and plants instead make it from glutamate through the C5 pathway, via glutamyl-tRNA†. Which one your host uses decides which precursor a haem-protein product competes for, which is a real consideration for the leghemoglobin of part 6 chapter 55.

Anaerobic operation: the split cycle

Without oxygen, *E. coli* does not run the cycle as a loop; it runs it as two branches. An oxidative branch carries oxaloacetate through citrate to α-ketoglutarate to supply glutamate-family precursors, and a reductive branch carries oxaloacetate through malate and fumarate to succinate, regenerating NAD^+ by using fumarate as an electron acceptor. That is the route the growth-coupled succinate design of part 11 chapter 39 exploited, and it is why succinate is an anaerobic product: the reductive branch consumes the NADH that anaerobic glycolysis cannot otherwise dispose of.

Respiration, oxygen, and the respiratory quotient

Complete oxidation of one glucose consumes six O_2 and releases six CO_2, so the **respiratory quotient**, RQ = CO_2 produced ÷ O_2 consumed, is 1.0 for a pure glucose oxidation. It is below 1 when the substrate is more reduced than glucose (fatty acids, 0.7), or when the product is reduced; and above 1 when part of the substrate is fermented, since fermentation releases CO_2 without consuming oxygen. RQ is a cheap, online physiological diagnostic from off-gas analysis, and a drift above 1 in a fed-batch is the first sign of overflow. In part 10's core model, 9.5 mmol of O_2 is consumed per 10 mmol of glucose while making 0.754 gDW per hour, which is **12.6 mmol of O_2 per gram of biomass formed, or 0.40 g O_2 per g** — the figure that sized the k_La requirement of part 11 chapter 44.

Overflow metabolism

When glucose uptake exceeds what respiration can handle, *E. coli* excretes acetate even in the presence of oxygen; *S. cerevisiae* makes ethanol (the Crabtree effect); mammalian cells make lactate (the Warburg effect). The older explanation was limited respiratory capacity. The modern one, from part 1 chapter 6, is allocation: respiration costs more protein per ATP than fermentation does, so above a threshold growth rate near 0.76 h^{-1}† the cell gains more ATP per gram of protein by fermenting part of the carbon. Part 11 chapter 39 reproduced the behaviour on the core model by adding a proteome budget with illustrative costs, and the onset of by-products as uptake rose was the result.

The three consequences stand: carbon leaves as by-product and yield falls; acetate inhibits growth, more at low pH where the weak acid enters the cell (part 2 chapter 14); and the operational answer is a fed-batch that holds the specific uptake rate below the threshold. Deleting the by-product pathway moves the problem, as part 1 chapter 6's worked example showed, rather than removing it.

Per turn	acetyl-CoA → 2 CO_2 + 3 NADH + $FADH_2$ + GTP
Per glucose	6 CO_2 + 10 NADH + 2 $FADH_2$ + 4 ATP (substrate-level)
Anaplerosis	PEP + CO_2 → oxaloacetate (no ATP); pyruvate + CO_2 + ATP → oxaloacetate
RQ	CO_2/O_2 = 1.0 for glucose oxidation; above 1 signals fermentation
Oxygen per biomass	12.6 mmol O_2 per gram formed in the part 10 model (0.40 g/g)
Overflow threshold	~0.76 h^{-1} in *E. coli*†; set by specific uptake rate

Common confusions

- **"The TCA cycle makes ATP."** Directly it makes one per turn; its main output is reducing equivalents and precursors.

- **"The carbons that enter as acetyl-CoA leave as CO_2 in the same turn."** They are released in later turns.
- **"Acetate means something went wrong."** It is the predicted response to uptake outrunning respiratory capacity.

Self-check — answer on paper first

1. A fed-batch shows an RQ of 1.15 and 2 g/L of acetate. What do you conclude and what do you change?

An RQ above 1 means CO_2 is being produced without a matching oxygen consumption, which is what fermentation does; together with acetate, it says the cells are running overflow. The change is the feed: lower the substrate feed rate so the specific uptake rate falls below the threshold, and the RQ should return to about 1.0 and the acetate be re-consumed. It is a quantitative diagnosis you can make online, without a sample, which is why the RQ is worth logging.

2. Why does withdrawing α-ketoglutarate for glutamate production require an anaplerotic reaction?

Because the cycle is closed: every α-ketoglutarate drawn off is a four-carbon unit that does not return as oxaloacetate to condense with the next acetyl-CoA, so the pool of cycle intermediates falls and the cycle slows. Replenishing oxaloacetate — by PEP carboxylase or pyruvate carboxylase — restores it at the cost of the carboxylation (and an ATP for the carboxylase), and that carboxylation, not the glutamate pathway itself, often sets the product yield.

3. A yeast and an *E. coli* both overflow. What differs in the mechanism and in the by-product?

The yeast makes ethanol, because it has pyruvate decarboxylase and alcohol dehydrogenase and is Crabtree-positive, so overflow begins at much lower glucose concentration; the bacterium makes acetate through phosphotransacetylase and acetate kinase, gaining an extra ATP as it does so. In both the driver is that fermentation yields more ATP per unit of protein invested than respiration when growth is fast; the by-product differs, and so does its toxicity and its value, which is why part 6 chapter 58 treated the Crabtree status of a yeast as a deciding host property.

What to open a book for. Lehninger chapters 16 to 19; Alberts is weaker here. For overflow metabolism and proteome allocation, Basan et al., Nature 528:99 (2015), plus a recent review since the quantitative account is recent. Part 11 chapter 39 of this module reproduces the effect on a model.

Chapter B7 · tracker week 7

Biosynthesis: what a gram of cells costs

Twelve precursors build everything. The useful number is not how many reactions that takes but how much glucose a gram of cells consumes — about 2.4 g — and where half of the carbon goes.

The cost of a gram, computed

Part 10's core model grows at 0.754 h^{-1} on 10 mmol of glucose per gram per hour, which means it uses 10/0.754 = **13.3 mmol of glucose to make one gram of biomass — 2.39 g of glucose**, a yield of 0.42 g/g. That figure comes out of the stoichiometry rather than being measured, and it is close to measured aerobic yields. The decomposition is the instructive part. A gram of cells contains about 48% carbon†, which is 40 mmol; as glucose that is 6.7 mmol. So **50% of the glucose carbon ends up in biomass and the other 50% leaves as CO_2**, for two reasons: carbon is lost during biosynthesis wherever a decarboxylation occurs (pyruvate dehydrogenase, the TCA cycle, the oxidative pentose phosphate pathway), and carbon is burned to make the ATP and NADPH that biosynthesis consumes.

That 50% is the baseline against which every product's yield is judged. A strain that makes a product from the same glucose competes with biomass for the same carbon, ATP and reducing power, which is the three-way split of part 1 chapter 1 — and it is why a growing cell can never convert more than about half its substrate carbon to cells, and why a product pathway must be cheaper than the biomass it displaces.

The twelve precursors, and the families they define

All biosynthesis begins from twelve compounds of central metabolism (part 1 chapter 7, figure 7.1): glucose-6-P, fructose-6-P, ribose-5-P, erythrose-4-P, glyceraldehyde-3-P, 3-phosphoglycerate, PEP, pyruvate, acetyl-CoA, α-ketoglutarate, succinyl-CoA and oxaloacetate. The core model's biomass reaction lists the demand in mmol per gram dry weight: acetyl-CoA is the largest at 3.7, pyruvate 2.8, oxaloacetate 1.8, α-ketoglutarate (with glutamate) the nitrogen entry, 3-phosphoglycerate 1.5, and so on†. In part 10's rounded version of that reaction the demands sum to 70.9 mmol of carbon per gram, more than the 40 in the biomass, because the extra carbon is lost on the way to the monomers — and that difference is the biosynthetic decarboxylation loss.

Family	Precursor	Amino acids	Notes
Glutamate	α-ketoglutarate	Glu, Gln, Pro, Arg	The nitrogen entry point; ammonium is assimilated here
Aspartate	Oxaloacetate	Asp, Asn, Lys, Thr, Met, Ile	Feedback-inhibited aspartokinases — the lysine and threonine engineering target
Aromatic	PEP + erythrose-4-P	Phe, Tyr, Trp	The shikimate pathway; competes with the PTS for PEP
Pyruvate	Pyruvate	Ala, Val, Leu	Valine and leucine feed the 2-ketoacid alcohol route of part 5 chapter 55
Serine	3-phosphoglycerate	Ser, Gly, Cys	Gly and Ser are the main one-carbon donors
Histidine	Ribose-5-P (via PRPP)	His	Shares PRPP with nucleotide synthesis

The costs differ per amino acid, and the published estimates of the ATP-equivalents spent range from roughly a dozen for the cheapest to several tens for the most expensive†, depending on whether the accounting includes the opportunity cost of the carbon and reducing power used. The engineering use is relative: the aromatic and branched-chain amino acids are expensive, glycine and alanine are cheap, and a protein product enriched in the expensive ones is correspondingly costlier to make.

Nitrogen and sulfur

Nitrogen enters through glutamate and glutamine. Glutamate dehydrogenase makes glutamate from α-ketoglutarate, ammonium and NADPH at a cost of no ATP but with low affinity for ammonium, so it works when ammonium is plentiful. When it is scarce the cell uses glutamine synthetase, which spends one ATP to attach ammonium to glutamate, then glutamate synthase (GOGAT) to transfer the amide nitrogen back to α-ketoglutarate; the pair has a much higher affinity and costs an ATP plus an NADPH per nitrogen. Every other amino group in the cell is then moved by transamination, using pyridoxal phosphate. Nitrogen assimilation is therefore tied to the TCA cycle through α-ketoglutarate, and a nitrogen-rich product draws on that pool (part 1 chapter 7).

Sulfur enters as sulfate and must be reduced to sulfide — an eight-electron reduction — before it can be put into cysteine. The cost is two ATP to activate sulfate to PAPS and four NADPH for the reductions†. That is an NADPH price of the same order as a fatty-acid elongation and is routinely omitted from back-of-envelope balances for sulfur-containing products such as glutathione or methionine. Methyl groups come from S-adenosylmethionine, made from methionine

and ATP, and each methylation leaves S-adenosylhomocysteine that must be recycled.

Lipids, nucleotides and the cell envelope

Fatty acids are built from acetyl-CoA. Acetyl-CoA carboxylase, a biotin enzyme, makes malonyl-CoA at a cost of one ATP; in bacteria a dissociated synthase (type II) then adds two carbons per cycle using two reductions — a ketoacyl reduction that uses NADPH and an enoyl reduction that *E. coli*'s FabI does with NADH †. A palmitate (C_{16}) therefore costs 8 acetyl-CoA, 7 ATP and 14 reducing equivalents, split about 7 NADPH and 7 NADH in *E. coli*; yeast's fatty acid synthase takes both from NADPH, giving the 14 NADPH of part 6 chapter 58, and adds the ATP-citrate lyase step so pays 15 ATP. **Phospholipids** add glycerol-3-phosphate and a headgroup, with CDP-activated intermediates.

Nucleotides start from phosphoribosyl pyrophosphate, made from ribose-5-phosphate and ATP. Purines are assembled on a ribose scaffold through inosine monophosphate, a ten-step pathway using glycine, glutamine, aspartate, one-carbon units from folate and a good deal of ATP; pyrimidines build the ring first and attach the ribose after, through orotate. Ribonucleotide reductase makes deoxyribonucleotides from ribonucleotides, and thymidylate synthase, which needs methylene-tetrahydrofolate, makes the thymidine — which is why folate antagonists are antimicrobials.

The **envelope** is the other large sink: peptidoglycan, whose synthesis is the target of the β-lactams and which is a substantial part of a Gram-positive cell; and lipopolysaccharide in Gram-negatives, which is the endotoxin that part 6 chapter 59 listed as an advantage of Gram-positive hosts for injectables. For product engineering the point is that the envelope is a fixed cost of growth, and a host with a smaller or simpler one has less to build.

Cost of a gram	13.3 mmol glucose = 2.39 g per gram of cells; yield 0.42 g/g
Where the carbon goes	48% C → 40 mmol → 6.7 mmol glucose-equivalent: 50% to biomass, 50% to CO_2
Precursor demand	70.9 mmol C/gDW in the core biomass reaction; the surplus over 40 is decarboxylation
Nitrogen	GDH: no ATP, low affinity · GS-GOGAT: 1 ATP + 1 NADPH per N, high affinity
Sulfate to sulfide	2 ATP + 4 NADPH (8 electrons)
Palmitate	8 acetyl-CoA + 7 ATP + 14 reducing equivalents (7 NADPH + 7 NADH in *E. coli*†; 14 NADPH in yeast)

Common confusions

- **"Biomass is cheap."** A gram costs 2.4 g of glucose and half the carbon is respired.
- **"Succinyl-CoA feeds haem."** In yeast and animals; *E. coli* uses glutamate through the C5 route.
- **"Ammonium assimilation is free."** It depends on the route; the high-affinity one costs an ATP and an NADPH per nitrogen.

Self-check — answer on paper first

1. Your strain makes a product at a carbon yield of 60%. Is that better or worse than making biomass, and why does the comparison matter?

Better: biomass converts only about 50% of the glucose carbon, so a product at 60% uses carbon more efficiently than growth does. The comparison matters because it is the three-way split of part 1 chapter 1 — carbon goes to cells, product or CO_2 — and a growth-coupled design (part 11 chapter 39) works precisely because product formation can be arranged to be the better use of carbon than growth. A product whose carbon yield is below 50% is competing at a disadvantage and needs its value to compensate.

2. A palmitate-derived product needs 14 NADPH per molecule. How many glucose does the reductant cost if it all comes from the oxidative pentose phosphate pathway, and what is the carbon penalty?

14/2 = 7 glucose pass through the oxidative branch (each gives 2 NADPH). In part 6 chapter 58 the corresponding figure is 1.17 glucose when the pentose phosphate pathway is fully cycled, because there the six carbons recycle through the non-oxidative branch giving 12 NADPH per glucose — the answer depends on which accounting you use, 2 per pass or 12 per cycle, and the accounting is the assumption. At 2 per pass the carbon penalty is 0.5 C per NADPH, 7 carbons for 14 NADPH.

3. Why is sulfate reduction usually omitted from yield estimates, and when does that matter?

Because the demand for sulfur in cell mass is small — a few per cent of the amino acids carry sulfur — so the NADPH cost is a minor part of the biomass budget. It matters when the product itself is sulfur-rich: glutathione, methionine, biotin, coenzyme A derivatives or iron–sulfur-dependent products. For glutathione, which has one cysteine per molecule, each molecule carries a sulfate-reduction cost of four NADPH and two ATP that a balance restricted to the glutamate and glycine steps will miss.

What to open a book for. Lehninger chapters 20 to 22 for the pathways; Stephanopoulos chapters 2 and 3 for the twelve-precursor yield arithmetic, which is the book you will keep using. The KEGG and MetaCyc pathway maps are the right reference for every enzyme step this chapter deliberately skips — read the map, then draw your own.

Molecular biology: replication, transcription, translation and their regulation

A gene's output is a product of a promoter, a ribosome binding site and two half-lives. Put numbers on each and you find that "strongly expressed" means about one per cent of the proteome — and that a consensus sequence alone predicts nothing.

Replication, in four facts that matter

Bacterial replication starts at one origin, *oriC*, where the initiator protein DnaA binds and opens the duplex; the helicase DnaB loads, and DNA polymerase III extends from RNA primers, continuously on the leading strand and in Okazaki fragments of one to two kilobases† on the lagging strand. Two forks proceed in opposite directions and meet at the terminus, where Tus protein blocks them. Four facts carry into engineering. The two forks together copy the genome in about 39 minutes (chapter B1). Initiation is regulated by cell mass, so a faster-growing cell initiates more often per generation and carries more than one set of origin-proximal genes (part 9 chapter 32). A plasmid has its own origin and its own copy-number control, independent of the chromosome (chapter B10). And replication errors are the main source of mutation, repaired by proofreading and mismatch repair at a residual rate near 10^{-10} per base per generation†.

Transcription: the promoter as a quantitative part

RNA polymerase is a core of five subunits ($\alpha_2\beta\beta'\omega$) that cannot start transcription alone. A sigma factor binds the core and recognises the promoter. The housekeeping factor σ^{70} recognises two hexamers, a −35 element near TTGACA and a −10 element near TATAAT, separated by a spacer of about 17 base pairs †; initiation melts the duplex around the −10 element, and the polymerase leaves the promoter after synthesising a short RNA. Termination is either intrinsic — a hairpin followed by a run of uridines destabilises the complex — or Rho-dependent, where a helicase chases the polymerase off. Alternative sigma factors redirect the same core to different promoter classes: σ^{38} in stationary phase and stress, σ^{32} for heat shock, σ^{54} for nitrogen-regulated genes, σ^{28} for flagella, σ^{24} for envelope stress†. The consequence for a production strain is the one part 1 chapter 8 gave: a promoter that is strong in exponential phase may be silent when the culture is stressed, which is when you want production.

A consensus match is not a promoter. How often would a six-base motif occur by chance in a 4.64 Mb genome? Counting both strands, an exact match to a given hexamer is expected 2,266 times; allowing one mismatch, which real promoters tolerate freely, the expected count is **43,047**. For a 12-base motif — a −35 and a −10 element together, ignoring the spacer — exact matches are expected 0.55 times. So a consensus-matching hexamer is almost always a coincidence, and only the combination of two elements at the right spacing, in the right context, with activators or repressors present, makes a promoter. It is the reason computational promoter prediction is unreliable, and the reason a measured, characterised part (part 8 chapter 26) is worth so much more than a sequence.

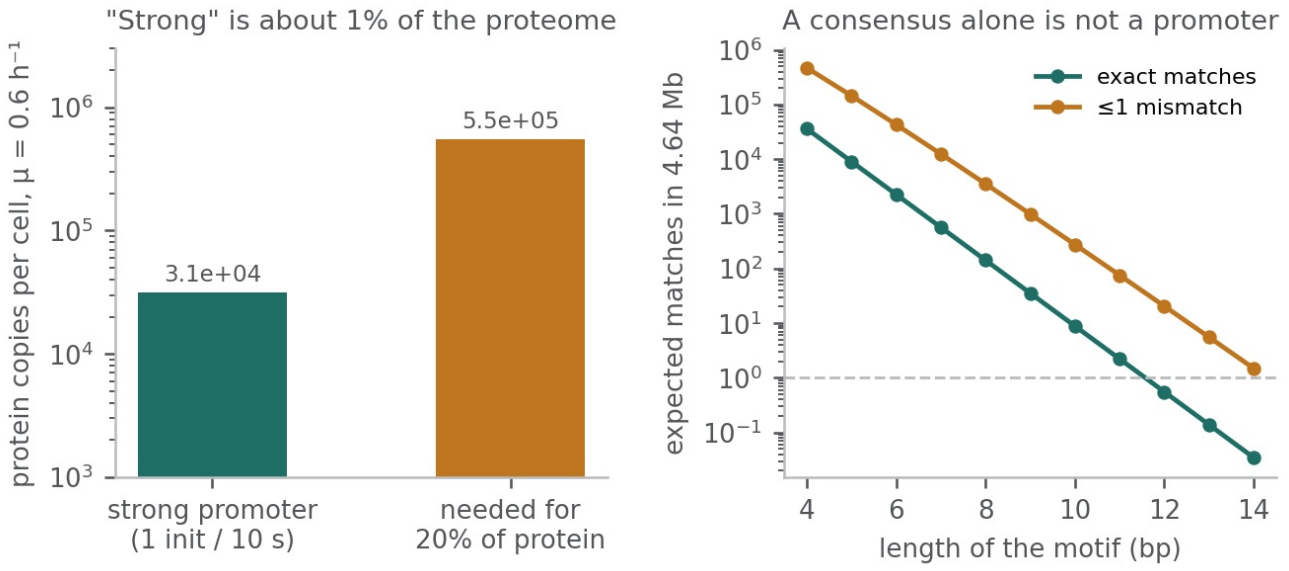


Figure B8.1 Left: protein copies per cell from a strong promoter at μ = 0.6 h⁻¹ (31,000, about 1.1% of the proteome) against the copies needed to make a protein 20% of the proteome (550,000). Right: expected random matches to a motif of the stated length in 4.64 Mb, exact and with one mismatch allowed. Computed for this module; the initiation rates and half-life are assumptions.

Translation: the second dial

The small ribosomal subunit binds the mRNA at the **Shine–Dalgarno sequence**, about AGGAGG, which pairs with the 3′ end of 16S rRNA five to nine bases upstream of the start codon†; initiator tRNA carrying formylmethionine is placed on AUG (the dominant start codon; GUG and UUG are used less often†) with the help of initiation factors IF1 to IF3, and the large subunit joins. Elongation cycles through aminoacyl-tRNA delivery by EF-Tu (spending GTP), peptide-bond formation, and translocation by EF-G (spending another). Termination occurs at a stop codon recognised by release factor RF1 (UAA, UAG) or RF2 (UAA, UGA), with RF3 assisting. Because initiation is usually rate-limiting, the ribosome binding site and the mRNA structure around the start codon set the protein level per mRNA — the independent dial of part 1 chapter 8, and the quantity the RBS calculator of part 8 chapter 28 predicts.

Codon choice matters at the margin. A handful of codons are rare in *E. coli* — AGA, AGG, AUA, CUA, CGA, CCC† — and a gene rich in them stalls ribosomes, depletes the matching tRNAs and can misincorporate or truncate. Whether to replace them is the "optimise or harmonise" question of part 8 chapter 30, and the rule remains to compare constructs and measure soluble active protein.

What a strong gene costs in numbers

Combine the two dials with the two half-lives, as part 1 chapter 8's expression equation did, and compute a concrete case. A strong promoter initiating once every 10 seconds makes 6 transcripts a minute; with an mRNA half-life of 3 minutes (decay constant 0.231 min^{-1}) the steady state is 6/0.231 = **26 mRNA copies**. A strong ribosome binding site loads a ribosome every 5 seconds, so each message yields 12 proteins a minute, and the cell makes 26 × 12 = 312 proteins a minute. A protein that is stable is lost only by dilution, at the growth rate: at $\mu = 0.6\ h^{-1}$ (0.01 min^{-1}) the steady state is 312/0.01 = **31,200 copies, 1.1% of the 2.8 million protein molecules**; at $\mu = 1.2\ h^{-1}$ it halves to 15,600, 0.56%.

Three conclusions. That is within a factor of a few of the most abundant proteins in the cell — the translation factor EF-Tu is about 5% of the protein† — so "strong promoter plus strong RBS" is already at the top of the natural range. Reaching 20% of the proteome needs 5.5×10^5 copies, **eighteen times more**, which a single chromosomal copy cannot supply and a high-copy plasmid with a very strong promoter can only approach by crowding out growth (chapter B9). And dilution works in the engineer's disfavour: the faster the cells grow, the lower the steady-state level of every stable protein, which is why a pathway expressed in exponential phase falls as a fraction of the proteome when growth accelerates.

Regulation by repressors and activators: the *lac* operon as the template

The operon *lacZYA* encodes β-galactosidase, a permease and an acetyltransferase from one promoter. The repressor LacI, a tetramer, binds an operator overlapping the promoter and blocks transcription; the inducer allolactose — or the non-metabolisable IPTG used in the lab — binds LacI and releases it. A second input comes from the catabolite activator protein CRP, which in complex with cyclic AMP binds upstream and recruits polymerase, and cyclic AMP is high only when glucose is absent. So the operon is expressed strongly only when lactose is present **and** glucose is not — an AND gate, in chemistry — as part 1 chapter 8 put it. Every inducible expression system in the module is a variation on this architecture, with the same properties: leak when the repressor is saturated, a Hill response set by cooperativity, and an inducer that must enter the cell.

Other layers act after transcription: RNase E degrades mRNA, with the half-life set partly by structure at the 5′ end; small RNAs pair with a transcript, usually with the chaperone Hfq, to change translation or decay; riboswitches in the leader fold differently on binding a metabolite and so regulate their own gene; and proteins are removed by proteases such as Lon and ClpXP, recruited by degradation tags like the ssrA sequence†. Each is a tool in part 8 chapter 27 and a source of context effects you must control.

σ^{70} promoter	−35 ≈ TTGACA, 17 bp, −10 ≈ TATAAT†; recognition by σ factor
Random matches, 4.64 Mb	hexamer exact 2,266; ≤1 mismatch 43,047; 12-mer exact 0.55
Shine–Dalgarno	≈ AGGAGG, 5–9 nt before AUG; initiation is rate-limiting
Expression budget	26 mRNA × 12 proteins/min ÷ 0.01 min^{-1} = 31,200 copies = 1.1% at μ 0.6 h^{-1}
To reach 20%	5.5×10^5 copies, 18× the strong-promoter case

Common confusions

- **"A sequence that matches the consensus is a promoter."** It is one of tens of thousands of matches.
- **"The promoter sets the expression level."** It sets transcription; the ribosome binding site and the two half-lives set the rest.

- **"A stronger promoter always gives more protein."** Not when growth slows — the level is production divided by (decay + dilution).

Self-check — answer on paper first

1. Your protein of interest is stable, expressed from a strong promoter, and cells grow at 0.3 h^{-1} in a nutrient-limited fed-batch. Estimate the level relative to the 0.6 h^{-1} case.

With production unchanged the steady-state level is production/dilution, so halving the growth rate doubles the level: 62,400 copies, 2.3% of the protein. The assumption that production stays the same is the weak one, since a slower-growing cell has fewer ribosomes and a different promoter activity (chapter B9) — so this is an upper bound, and the real level lies between the two cases. It does show why a protein that looks well expressed in a rich-medium shake flask can behave very differently in a slow fed-batch.

2. Two genes in one operon give protein ratios of 1:8 although they share a promoter. Why?

They share one mRNA, so transcription cannot differ; what differs is translation initiation, set by each gene's own ribosome binding site and local mRNA structure, and possibly differential mRNA decay along the transcript. A 1:8 ratio is a ribosome binding site difference of about eightfold, well within what the RBS calculator spans, which is how pathway stoichiometry is tuned inside one operon (part 1 chapter 8's worked example).

3. Why does the AND-gate logic of the lac operon matter for a production process that uses IPTG?

IPTG bypasses the lactose arm — it releases the repressor regardless of the sugar supply — but the CRP–cAMP arm still operates, so expression from a lac-type promoter can be lower in glucose-grown cultures than in glycerol- or lactose-grown ones even at saturating IPTG. A fed-batch on glucose can therefore show weaker induction than the same strain on another carbon source, and the cure is either a promoter without CRP dependence or a carbon source that does not repress.

What to open a book for. Alberts chapters 4 to 8 selectively; Lehninger chapters 24 to 28 for the chemistry of the information pathways. Snyder & Champness for bacterial gene regulation written around experiments, which is the better route into the lac operon than any mechanism-first chapter.

System-level regulation and the growth laws

A cell with a fixed protein budget grows at a rate set by how much of it is ribosomes. Every protein you add that does not help growth is paid for out of that budget — and the price, computed here, is about 1.8% of growth rate for each 1% of proteome.

Global preferences: catabolite repression

Given two carbon sources, *E. coli* uses the better one first and represses the machinery for the other, giving the diauxic growth curve with a lag between phases. The mechanism runs through the phosphotransferase system. While glucose is being imported, one component of the PTS, $EIIA^{Glc}$, is mostly dephosphorylated; in that state it fails to activate adenylate cyclase, so cyclic AMP is low and CRP–cAMP cannot activate the operons for other sugars, and it also blocks the lactose permease directly (inducer exclusion)†. When glucose is exhausted, $EIIA^{Glc}$ becomes phosphorylated, cyclic AMP rises, and the alternative operons switch on after a lag while the enzymes are made. The engineering consequence from part 1 chapter 9 stands: a mixed feedstock is consumed sequentially, and simultaneous utilisation is a project, not a default.

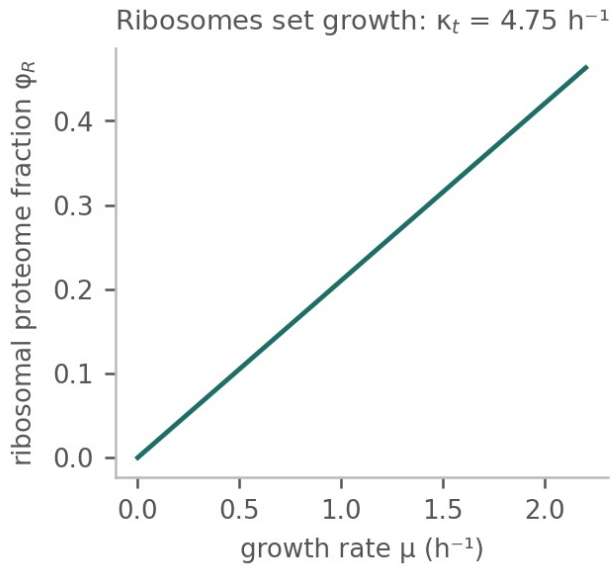
The proteome is a budget, and ribosomes set the growth rate

Chapter B1 computed the number of ribosomes a given growth rate requires. Now invert it into the relation that organises the field. If a fraction φ_R of the protein is ribosomal protein, and each ribosome (7,500 residues of protein per ribosome†) elongates at k = 18 aa/s with a fraction f = 0.55 of them active, the protein synthesis rate per unit protein is $(k \cdot f/L_R) \cdot \varphi_R$, which is the growth rate:

```
μ = κ_t · φ_R,      κ_t = k·f/L_R = 18 × 0.55 / 7,500 × 3,600 = 4.75 h⁻¹
```

Growth rate μ (h^{-1})	Growth-dependent ribosomal proteome fraction $\varphi_R = \mu/\kappa_t$
0.3	0.06
0.6	0.13
1.0	0.21
1.5	0.32
2.0	0.42

A linear relation of ribosome content to growth rate is exactly what is measured, with a slope in the region of 4.5 h^{-1} for the translational capacity†; my computed 4.75 h^{-1} is an order-of-magnitude check on the model, built from three recalled constants, not a measurement. What it says is the point: **the doubling time is set by how much of the proteome is devoted to making proteins**. At 2 h^{-1} over 40% of the protein is ribosomal; there is no room for a large uncommitted expression load.



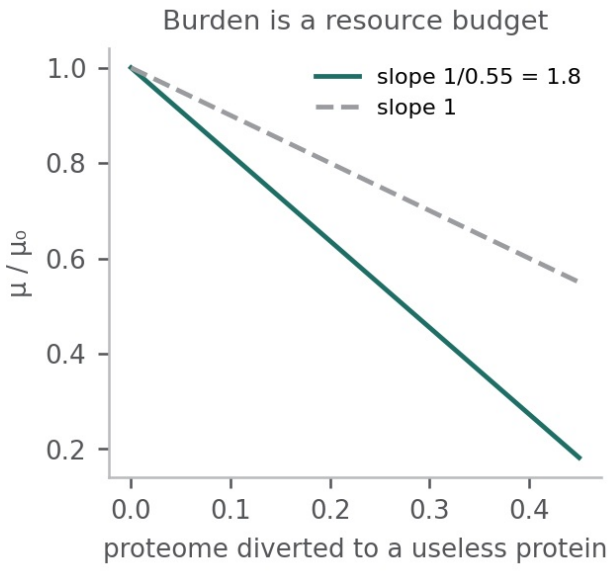


Figure B9.1 Left: the growth-dependent ribosomal proteome fraction against growth rate, from $\mu = \kappa_t \varphi_R$. Right: the fall in growth rate when a fraction of the proteome is diverted to a useless protein, for a maximum diversion $\varphi_{max} = 0.55$ (slope 1.8) and for slope 1. Computed for this module from the stated constants.

The price of burden, in numbers

Because the proteome is a fixed budget, expressing a protein that does not contribute to growth displaces the growth-supporting sectors and slows the cell. The simplest form, which fits measurements of cells expressing a useless protein†, is $\mu/\mu_0 = 1 - \varphi_X/\varphi_{max}$, with φ_{max} about 0.55†, so the slope is 1/0.55 = 1.8. Computed:

Proteome diverted, φ_X	μ/μ_0	Growth lost
1.1% (the strong gene of chapter B8)	0.98	2%
5%	0.91	9%
10%	0.82	18%
20%	0.64	36%
30%	0.45	55%

So **a single very strong gene costs about 2% of the growth rate, a pathway taking a fifth of the proteome costs a third, and one taking 30% halves it**. This is the quantitative form of part 1 chapter 9's "burden is measurable". It also connects to part 8 chapter 28, where the library model swept a burden slope and found the best combination changing at 2.4 and at 3.6: a slope near 1.8 puts a heavy pathway just below the first crossover, so the expression-everywhere design is on the edge of being the wrong one — the library is how you find out on which side your pathway sits. The burden depends on the medium, as chapter B8's discussion of slow growth implies; the slope is steeper in poorer media†.

Stress responses and the stringent response

The stringent response is triggered when amino acids run short. An uncharged tRNA entering the ribosome activates RelA, which synthesises the alarmone ppGpp (and SpoT makes and degrades it); ppGpp, with its partner DksA, binds RNA polymerase and shuts down transcription of ribosomal RNA and tRNA while raising that of amino acid biosynthesis genes. The cell stops investing in growth and shifts to maintenance. Anything that drains an amino-acid pool — overexpressing a protein rich in one residue, a toxic intermediate, a starved fed-batch — can fire it, and production falls with growth (part 1 chapter 9).

The other global responses to know by name: the heat-shock response through σ^{32}, which induces chaperones and proteases and is triggered by misfolded protein — so heavy overexpression induces it; the general stress response through σ^{38} in stationary phase; the envelope stress response through σ^{24} and the Cpx system; the SOS response to DNA damage, where RecA stimulates cleavage of the LexA repressor and raises mutation rates; and oxidative stress through OxyR and SoxRS. A strain that underperforms for no pathway reason is often showing one of them.

Sensing: two-component systems and the global regulators

A two-component system is a sensor histidine kinase in the membrane and a response regulator in the cytoplasm. The sensor autophosphorylates on a histidine when it detects its signal and transfers the phosphoryl group to an aspartate on the regulator, which then binds DNA — EnvZ and OmpR for osmolarity, PhoR and PhoB for phosphate, NtrB and NtrC for nitrogen. The others to know: Fnr, which senses oxygen through an iron–sulfur cluster and switches on anaerobic genes; ArcAB, which senses the quinone redox state; Fur, for iron; and the quorum-sensing systems, where cells secrete a small diffusible signal — an acyl-homoserine lactone in many Gram-negatives, a peptide in Gram-positives — and respond above a density threshold, which is what part 8 chapter 27 used as a clock.

When a strain underperforms: the diagnostic

The measurements to take before touching the pathway, as in part 1 chapter 9, now with numbers to interpret them: growth rate with and without induction, and compare it with $\mu/\mu_0 = 1 - \varphi_X/0.55$ to infer how much proteome you are using; by-product formation and the respiratory quotient (chapter B6), to see whether the specific uptake rate has crossed the overflow threshold; a carbon balance, to find where the carbon went; and a stress reporter, to see whether a global response has fired. If the growth rate fell by more than the expression level predicts, a global response or a toxic intermediate is the likelier cause than a pathway defect.

Growth law	$\mu = \kappa_t \varphi_R$, $\kappa_t = kf/L_R = 4.75\ h^{-1}$ (computed); ~4.5 h^{-1} measured†
Ribosome fraction	0.13 at 0.6 h^{-1}; 0.21 at 1.0; 0.42 at 2.0

Burden	$\mu/\mu_0 = 1 - \varphi_X/\varphi_{max}$, $\varphi_{max} \approx 0.55$ → slope 1.8
Price of expression	1.1% of proteome → 2% growth; 20% → 36%; 30% → 55%
Stringent response	uncharged tRNA → RelA → ppGpp → rRNA transcription down, amino-acid synthesis up

Common confusions

- **"Burden is vague."** It is a budget with a measured slope near 1.8 per unit of proteome.
- **"A mixed sugar medium is consumed together."** Usually sequentially, because CRP–cAMP and inducer exclusion enforce the order.
- **"The stringent response is a starvation curiosity."** A heavy overexpression can trigger it and production falls with growth.

Self-check — answer on paper first

1. A pathway takes 12% of the proteome. How much slower do the cells grow, and what would you do if that is too much?

$\mu/\mu_0 = 1 - 0.12/0.55 = 0.78$, so 22% slower. If that is unacceptable, the options of part 1 chapter 9 apply: stage the process so growth finishes before expression begins, which keeps both; lower the expression of the enzymes with low control coefficients (part 2 chapter 15), since those cost proteome and buy little flux; or move the least-needed enzymes to a second strain (chapter S4), which halves each one's burden.

2. Why does the ribosome fraction have to rise faster than linearly in the number of ribosomes when growth speeds up?

Two effects multiply. The protein synthesis rate required is μ times the protein content, so it rises with μ; and at higher μ the cell is larger and holds more protein, so the content rises too. Chapter B1's calculation showed a 3.3-fold growth rate increase needing a 6.7-fold ribosome increase; the proteome fraction — the ratio of ribosomal to total protein — rises linearly with μ, but the absolute count rises faster, and both are what the growth law and the ribosome count say.

3. Your IPTG-induced strain stops growing 30 minutes after induction. Name the first three things you check.

The proteome fraction the protein takes, by gel or by a ratio of its band to total protein — if it exceeds 25 to 30%, the burden alone explains it; the ppGpp level or a stringent-response reporter, since a protein rich in a scarce amino acid can drain it; and whether the protein is toxic in its own right, by testing an inactive point mutant at the same expression level, which separates burden from activity. The heat-shock response is a fourth, with misfolded protein as its trigger.

What to open a book for. Scott et al., Science 330:1099 (2010), which is the paper behind the growth laws and is short and readable; Basan et al., Nature 528:99 (2015) for the overflow link; Alberts chapter 7 for the regulatory mechanisms, and a recent review of proteome allocation for the quantitative account.

Chapter B10 · tracker week 10

Microbial genetics and microbiology: growth, mutation, gene transfer, sterility

Mutations arise before you select, not because you select — and that one result, derived below from a simulation with a variance-to-mean ratio of 649, is why every large culture already contains the variant that will take it over.

Growth, yield and maintenance

Batch growth has a lag, an exponential phase in which biomass follows $X = X_0 e^{\mu t}$, a stationary phase and decline; the doubling time is $t_d = \ln 2/\mu$. The dependence of μ on substrate is Monod's: $\mu = \mu_{max} S/(K_s + S)$, a Michaelis–Menten form with $\mu = \mu_{max}/2$ at $S = K_s$. In a chemostat at steady state $\mu = D$, so the residual substrate is $S = K_s D/(\mu_{max} - D)$; with $\mu_{max} = 0.9$ h⁻¹ and $K_s = 0.5$ g/L, a dilution rate of 0.3 h⁻¹ leaves 0.25 g/L and 0.8 h⁻¹ leaves 4.0 g/L — the sharp rise that precedes washout.

The observed biomass yield falls at low growth rates because the cell spends part of its substrate on maintenance. Pirt's relation is $1/Y_{obs} = 1/Y_{true} + m/\mu$. With an illustrative true yield of 0.50 g/g and a maintenance of 0.058 g glucose per g biomass per hour — which is the core model's 8.4 mmol ATP per g per h at about 26 ATP per glucose — the observed yield is:

Growth rate μ	Observed yield
0.60 h⁻¹	0.48 g/g
0.10 h⁻¹	0.39 g/g
0.05 h⁻¹	0.32 g/g
0.02 h⁻¹	0.20 g/g

That table is the quantitative reason a very slow fed-batch is expensive, which part 11 chapter 44's feeding discussion balanced against overflow: below a growth rate of about 0.1 h⁻¹ the maintenance term takes a growing share of the carbon, and a product that is not growth-coupled competes with it.

Counting, and the 30 to 300 rule

Three measurements of "how many cells" disagree for reasons part 1 chapter 10 gave: optical density tracks biomass and cell size, colony counts measure cells able to grow on that medium, and dry weight is what a balance needs. A colony count is a Poisson variable, so its relative uncertainty is $1/\sqrt{N}$:

Colonies counted	Coefficient of variation
10	32%
30	18%
100	10%
300	5.8%
1,000	3.2%

The conventional 30 to 300 counting window is the compromise between that statistical floor at the bottom and the crowding that makes colonies merge at the top. The arithmetic of a plate count is: 150 colonies from 0.1 mL of a 10^{-6} dilution is $150/(0.1 \times 10^{-6}) =$ **1.5×10^9 CFU/mL**, with an uncertainty of 8% from the count alone — and larger once pipetting and dilution error are added, which is why replicate plates at two adjacent dilutions are standard.

Mutation, and why it is always ahead of you

Do mutations arise because a selective condition calls them forth, or do they arise at random before it, and the selection merely finds them? Luria and Delbrück answered by noting that the two models predict different *distributions* of mutant counts across parallel cultures. If mutations were induced at the moment of selection, each culture would give a Poisson number of mutants, with variance equal to the mean. If they arise at random during growth, a mutation that happens early produces a large clone — a jackpot — and the variance greatly exceeds the mean.

I simulated it: 4,000 cultures each grown from 10^3 cells to 1.3×10^8 over 17 generations, with a mutation rate of 2×10^{-9} per cell per division.

	Mean mutants	Variance / mean	Largest culture
Random mutation during growth (simulated)	1.70	**649**	**2,048**
Induced at selection (Poisson, same mean)	1.70	1.0	8

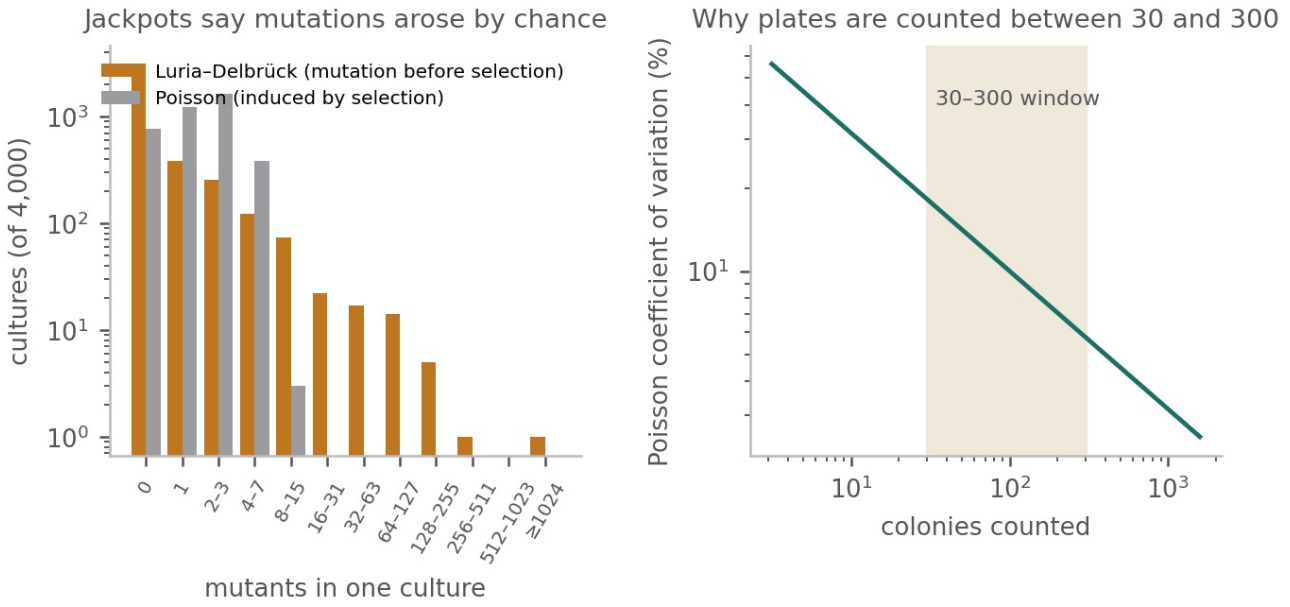


Figure B10.1 Left: the distribution of mutant counts across 4,000 simulated cultures under the Luria–Delbrück model against a Poisson distribution with the same mean. Right: the Poisson coefficient of variation of a colony count, with the 30–300 counting window. Computed for this module.

The difference is enormous, and it is the result: **a variance-to-mean ratio of 649 against 1.0, and a jackpot of 2,048 mutants where a Poisson distribution gives at most 8**. The experiment's data match the first. The simulation also gives the practical estimator: the fraction of cultures with no mutants is $p_0 = e^{-m}$, where $m = \mu N_f$ is the expected number of mutation events, so $\mu = -\ln p_0 / N_f$. Here $p_0 = 0.777$ and $N_f = 1.31 \times 10^8$ give $\mu = 1.9 \times 10^{-9}$, against the true 2.0×10^{-9} — the "p_0 method" recovers the rate from the number of empty cultures alone.

The engineering consequence is the one part 9 chapter 32 stated as instability. A loss-of-function mutation in a given gene occurs at roughly 10^{-10} per base per generation × 1,000 bases × about half of the mutations being inactivating† ≈ 5×10^{-8} per generation. A 1,000 L fermenter at 10^{10} cells/mL holds 10^{16} cells, so about **5×10^8 new loss-of-function mutants of any one gene appear every generation**. Whether they take over depends on their growth advantage — a pathway-inactivating mutation that removes burden has one — and on the number of generations. That is the reason scale-up adds a genetics problem, and the reason designs that couple production to growth (part 11 chapter 39) are valuable: they make the inactivating mutant lose the race.

Moving DNA between cells

Route	What happens	Engineering relevance
Transformation	Uptake of free DNA — natural competence in some species; $CaCl_2$/heat shock or electroporation in *E. coli*	Efficiency in transformants per µg decides whether a library is possible (part 7 chapter 25)
Conjugation	DNA transferred from donor to recipient through a pilus; needs an origin of transfer and a helper that supplies the transfer genes	The standard route into *Streptomyces* and other resistant hosts (part 8 chapter 31)
Transduction	A phage carries host DNA — any fragment in generalised transduction (P1), neighbouring genes in specialised (λ)	Moving a marked allele between strains

Plasmids, markers and compatibility

A plasmid replicates from its own origin, and the origin sets its copy number†: pUC-type high-copy ColE1 derivatives at several hundred per cell, pBR322 at 15 to 20, p15A at 10 to 12, pSC101 at about five, and F-factor bacterial artificial chromosomes at one or two. The ColE1 family controls copy number with an antisense RNA, RNA I, that pairs with the RNA II primer and prevents it from priming replication — a natural regulatory circuit of exactly the kind chapter B8 described. Two plasmids with the same type of origin are **incompatible** and one is lost, so co-expression plasmids use different families — a ColE1, a p15A and a pSC101 can coexist.

Marker	Gene and mechanism	Note
Ampicillin	*bla*, a β-lactamase that degrades the drug	Secreted, so it clears the drug around a colony and lets satellite colonies grow
Kanamycin	*aph*, an aminoglycoside phosphotransferase	Stable and widely used
Chloramphenicol	*cat*, an acetyltransferase	Bacteriostatic; works at lower plasmid copy numbers
Tetracycline	*tetA*, an efflux pump	The pump runs on the proton motive force (chapter B4), so it costs energy and burden
Spectinomycin	*aadA*, an adenylyltransferase	Common for integrative constructs

None of these belongs in a production process, so the standard replacements are auxotrophic complementation — a plasmid that restores a gene deleted from the chromosome — and integration. Counterselection markers, which kill cells that still carry them, such as *sacB* (lethal on sucrose) or *ccdB*, are what make marker removal and two-step genome editing work.

Sterility, computed

Killing by heat is first order, so the survivors fall as $N = N_0 \cdot 10^{-t/D}$, with D, the decimal reduction time, the minutes to kill 90%. The most heat-resistant contaminants are bacterial spores, with D_{121} of the order of 1.5 minutes for *Geobacillus stearothermophilus*†. A 100 L medium containing 10^6 spores per mL has 10^{11} spores. To make the probability of a surviving spore only 10^{-3} per batch needs a reduction of 10^{14}, **14 D**, which at 121 °C is 14 × 1.5 = **21 minutes**. The dependence on temperature is exponential: with a z-value — the temperature rise that cuts D tenfold — of 10 °C†, 14 D takes 21 minutes at 121 °C, 2.1 minutes at 131 °C and 0.21 minutes at 141 °C.

The reason plants use short high-temperature sterilisation is also computable. The nutrient damage — the browning and degradation of sugars and vitamins — has a much larger z-value, of the order of 30 °C †, so it rises with temperature far more slowly than spore killing does. Raising the temperature from 121 to 131 °C cuts the exposure time to a tenth while the damage rate rises only by $10^{10/30} = 2.15$, so the total damage falls to 0.1 × 2.15 = 0.215 — **4.6 times less** — for the same sterility. That asymmetry is the whole design logic of a continuous high-temperature, short-time steriliser.

Heat-labile components are filter-sterilised through a 0.2 µm membrane, which removes bacteria but not viruses; phage can pass, which is why phage contamination in a plant is managed by prevention and strain resistance and not by filtration alone (part 1 chapter 10). Aseptic technique itself, and the biosafety framework that surrounds it, are in part 9 chapter 32.

Monod	$\mu = \mu_{max} S/(K_s + S)$; chemostat $S = K_s D/(\mu_{max} - D)$
Maintenance	$1/Y_{obs} = 1/Y_{true} + m/\mu$; 0.48 → 0.20 g/g from µ 0.6 to 0.02
Counting	$CV = 1/\sqrt{N}$; 18% at 30 colonies, 5.8% at 300
Luria–Delbrück	variance/mean 649, jackpot 2,048 against Poisson 1.0 and 8; $\mu = -\ln p_0/N_f$
Mutants at scale	5×10^8 new inactivating mutants per generation in 10^{16} cells
Thermal death	$N = N_0 10^{-t/D}$; 14 D = 21 min at 121 °C; HTST gives 4.6× less nutrient damage

Common confusions

- **"Resistance appears because we applied the antibiotic."** The variants were already there; selection found them.
- **"A plate with 12 colonies is a good count."** Its uncertainty is 29%; count between 30 and 300.
- **"Longer sterilisation is always safer."** It is safer for spores and worse for the medium; the high-temperature short-time optimum exists because the two have different z-values.

Self-check — answer on paper first

1. Of 30 parallel cultures, 24 contain no resistant mutants after 10^8 cells each. Estimate the mutation rate.

$p_0 = 24/30 = 0.8$, so $m = -\ln 0.8 = 0.223$ and $\mu = 0.223/10^8 = 2.2 \times 10^{-9}$ per cell per division. With only 30 cultures the uncertainty is large — the standard error of p_0 is $\sqrt{(0.8 \times 0.2/30)} = 0.073$, which moves µ by roughly ±30% — so report the interval, and use 50 to 100 cultures if the rate matters. If the other six cultures show a few mutants and one shows hundreds, the jackpot confirms the mutations were not induced.

2. A 1,000 L fermentation needs a 10^{-4} chance of a surviving spore from 10^8 spores per batch. How many D-values, and what time at 121 °C with D = 1.5 min?

$\log_{10}(10^8/10^{-4}) = 12$ D; 12 × 1.5 = 18 minutes. Because the medium takes time to heat and cool, the lethality

accumulated during the ramps counts too, and the real hold time is usually set by integrating the lethal rate over the whole temperature profile rather than by the hold alone.

3. Why can two plasmids with ColE1-type origins not be maintained together, and what do you do?

They share the same regulatory RNA I and priming mechanism, so each controls the other's copy number as well as its own, and one is gradually diluted out as the cell enforces a single plasmid copy budget — incompatibility. Choose plasmids from different families, such as a ColE1 and a p15A origin, with different antibiotic markers; or put one of the genes on the chromosome by integration, which removes the compatibility question and the need for the second antibiotic.

What to open a book for. Brock *Biology of Microorganisms* is the reference for this week and the book most worth having for phase 1. MIT OCW 7.13 Experimental Microbial Genetics for the genetics in practice, and Snyder & Champness for selection design. Doran for sterilisation design and the thermal death arithmetic done as engineering problems.

Glossary

Each definition is original wording of a standard concept. Terms are those the tracker assumes.

Term	Meaning
Allosteric	Regulated by a ligand binding at a site other than the active site, changing activity
Anaplerosis	Reactions that replenish cycle intermediates drawn off for biosynthesis
Antiporter / symporter	Transporters that move two substances in opposite / the same direction
Auxotroph	A strain unable to make a compound and therefore needing it supplied
Catabolite repression	Repression of the use of secondary carbon sources while a preferred one is present
Chaperone	A protein that assists folding and prevents aggregation without becoming part of the product
Chemostat	Continuous culture at constant feed and removal; $\mu = D$ at steady state
Codon	Three nucleotides specifying an amino acid or a stop
Cofactor / coenzyme	A non-protein helper of an enzyme; a coenzyme is an organic one, often a vitamin derivative
Conjugation	Transfer of DNA between cells in contact
Cooperativity	Binding at one site changing affinity at others, giving a sigmoidal response
CRP (CAP)	cAMP receptor protein, a global activator of genes for alternative carbon sources
Crabtree effect	Fermentation to ethanol by yeast despite oxygen, at high sugar
D-value	Time at a temperature to kill 90% of a population
Dilution rate	Feed flow divided by culture volume; equals μ in a chemostat
Doubling time	$\ln 2/\mu$
EC number	Four-part code classifying an enzyme by the reaction it catalyses
Electron transport chain	Membrane complexes passing electrons to a terminal acceptor while pumping protons
Entner–Doudoroff pathway	An alternative glycolysis giving 1 ATP, 1 NADH and 1 NADPH per glucose
Fermentation	Energy metabolism without an external terminal electron acceptor
Flux	Rate of conversion through a reaction or pathway, per unit biomass or volume
Gram-negative / positive	Two envelope types: outer membrane plus thin wall, or one membrane plus thick wall
Hill coefficient	An empirical steepness of a sigmoidal response; not a count of sites
Hydrophobic effect	The tendency of non-polar groups to cluster to release ordered water; the main folding force
Inducer	A molecule that turns a gene on, usually by releasing a repressor
Isoelectric point (pI)	The pH at which a molecule has zero net charge
Isozyme	Different proteins catalysing the same reaction
k_{cat}	Turnover number: maximum rate per active site
K_M	Substrate concentration at half-maximal rate; $(k_{-1} + k_{cat})/k_1$
Lag phase	The adjustment period before exponential growth
Lysogeny	A phage integrating into the host genome instead of lysing it
Maintenance	Energy spent to stay alive, independent of growth
Mass-action ratio (Q)	Product of product concentrations over substrates; compared with K_{eq}
Missense / nonsense / synonymous	Mutations changing an amino acid / creating a stop / leaving the amino acid unchanged
Monod equation	$\mu = \mu_{max} S/(K_s + S)$
Operator	The DNA site where a repressor binds
Operon	Genes transcribed together from one promoter
Overflow metabolism	Excretion of fermentation products despite sufficient oxygen, when uptake outruns respiration

Oxidative phosphorylation	ATP synthesis driven by the proton motive force from respiration
P/O ratio	ATP made per pair of electrons passed to oxygen
Peptide bond	The amide link between amino acids; planar
Plasmid	An extrachromosomal replicon with its own origin
Polycistronic mRNA	One transcript encoding several proteins
Promoter	The DNA site where RNA polymerase starts transcription
Proton motive force (Δp)	Electrochemical proton gradient across a membrane, $\Delta\psi - 61.5 \cdot \Delta pH$ mV
PTS	Phosphotransferase system: imports and phosphorylates sugars using PEP
Quorum sensing	Density-dependent gene regulation through a secreted signal
Redox potential ($E^{\circ\prime}$)	A measure of the tendency of a couple to accept electrons
Regulon	Genes at different locations controlled by one regulator
Repressor	A protein that blocks transcription by binding an operator
Respiratory quotient	CO_2 produced over O_2 consumed
Ribosome binding site	The mRNA sequence, including Shine–Dalgarno, where translation starts
Riboswitch	An mRNA element that changes fold on binding a metabolite
Sigma factor	A subunit directing RNA polymerase to a class of promoters
Stringent response	Amino-acid starvation response via ppGpp that shuts down ribosome synthesis
Substrate-level phosphorylation	ATP made directly by a reaction, without a gradient
Terminator	A sequence ending transcription
Transduction	DNA transfer by a phage
Transformation	Uptake of free DNA by a cell
Transhydrogenase	An enzyme converting NADH and NADPH to each other
Two-component system	A sensor kinase and response regulator pair that transmits a signal by phosphorylation
Vitamin	An organic nutrient that an organism cannot make and needs for a cofactor
Wobble	Flexibility in pairing at the third codon position
Yield coefficient	Product or biomass formed per substrate consumed
z-value	The temperature rise that reduces D tenfold

Recalled constants to verify

Every value marked † in this part was recalled from standard references rather than computed. They are listed here with where to check each. A computed result that depends on one of them inherits its uncertainty, and chapters B1, B4, B8 and B9 are the ones most exposed.

Constant	Value used	Where to verify
Dry weight per cell, protein fraction, composition	0.3 pg; 55% protein; 20% RNA; 9% lipid; 3% DNA	BioNumbers; Neidhardt, *Escherichia coli and Salmonella*
Genome size, gene count, coding fraction	4.64 Mb; ~4,300; ~88%	EcoCyc; NCBI genome record
Fork speed, transcription and translation rates	1,000 nt/s; 40–80 nt/s; 10–20 aa/s	BioNumbers; Alberts chapters 5–6
mRNA half-life; average protein mass	3 min; 36 kDa	BioNumbers; proteome-wide measurements
ATP cost per peptide bond; ribosome counts	~4; ~10^4 at 0.6 h^{-1}, ~7 × 10^4 at 2 h^{-1}	Lehninger chapter 27; Bremer & Dennis, EcoSal
Amino-acid pK_a values	D 3.9, E 4.2, H 6.0, C 8.3, Y 10.1, K 10.5, R 12.5; termini 2.0, 9.0	Lehninger chapter 3
Helix geometry; folding stability range	3.6/turn, 1.5 Å; 20–60 kJ/mol	Lehninger chapter 4; Alberts chapter 3
Median k_{cat}, K_M; diffusion limit	~10 s^{-1}; ~100 µM; 10^8–10^9 $M^{-1}s^{-1}$	Bar-Even et al., Biochemistry 50:4402 (2011)
Vitamins and groups carried; EC classes	as tabulated	Lehninger chapter 6; the IUBMB enzyme list
Standard potentials	NAD −0.320, NADP −0.324, O_2 +0.816, etc.	Lehninger chapter 19
ATP $\Delta G^{\circ\prime}$ and cell concentrations	−30.5 kJ/mol; 3, 0.5, 10 mM	BioNumbers; Park et al., Nat. Chem. Biol. 12:482 (2016)
Δψ and pH values; c-ring sizes	−140 mV; 7.6/7.0; c_8, c_{10}, c_{12}	Nicholls & Ferguson; Lehninger chapter 19
E. coli respiratory chain variants; H^+/e^-	NDH-1, NDH-2, cytochromes bo_3 and bd	Unden & Bongaerts; EcoCyc
ED yield; PFK regulation; PTS role	1 ATP, 1 NADH, 1 NADPH; ADP/PEP	Lehninger chapter 14; EcoCyc
Biomass NADPH; carbon content	13.03 mmol/gDW; 48%	BiGG core model biomass reaction; Neidhardt
Anaplerotic enzymes by host; haem C5 route	PYC absent in *E. coli*; C5 in most bacteria	EcoCyc; MetaCyc
Amino-acid cost range; sulfate reduction; FabI cofactor	~12–75 ATP-eq.; 2 ATP + 4 NADPH; NADH	Akashi & Gojobori, PNAS (2002); MetaCyc
EF-Tu abundance; rare codons	~5% of protein; AGA AGG AUA CUA CGA CCC	Ecomics/BioNumbers; codon-usage tables
Promoter and SD consensus; start-codon usage	TTGACA/TATAAT/17 bp; AGGAGG; AUG ≫ GUG > UUG	Alberts chapter 6; Snyder & Champness
Growth-law constants	7,500 aa/ribosome; φ_{max} 0.55; κ_t ~4.5 h^{-1}	Scott et al., Science 330:1099 (2010)
Overflow threshold	0.76 h^{-1}	Basan et al., Nature 528:99 (2015)
Mutation rate; inactivation fraction	~10^{-10}/bp/gen; ~50% of mutations	Lee et al., PNAS 109:E2774 (2012)
Plasmid copy numbers	pUC ~500; pBR322 15–20; p15A 10–12; pSC101 ~5	Addgene plasmid reference
D and z values	D_{121} 1.5 min; z 10 °C (spores); z 30 °C (nutrients)	Doran; a food-microbiology reference

Closing notes to part 14

What this part adds, and what it still cannot

Ten chapters, 39 pages, about 1.7 times part 1's density per week. It supplies the quantitative core that the textbooks leave you to assemble: the codon-position structure of point mutations, the ribosome budget and what a protein costs, the ionisation arithmetic behind a His-tag purification, the barrier arithmetic of catalysis, the regime-dependence of an assay, the proton motive force and the minimum H^+/ATP, the decomposition of an ATP yield into its assumptions, the NADPH bound on the pentose phosphate pathway, the carbon cost of a gram of cells, the 1% of proteome that a strong gene occupies, the growth law and the price of burden, the Luria–Delbrück statistics, and thermal death. All of it is computed, with the recalled inputs marked.

What it still does not replace is the textbook's work: the mechanisms drawn atom by atom, the structures, the full pathway maps with every enzyme and cofactor, the experimental history that makes the facts stick. For weeks 1 to 10 you still need Lehninger for chapters 3 to 6, 13 to 22 and 24 to 28, Alberts for the cell and the information pathways, and Brock for week 10. This part is meant to be read first, to know what matters in each, and then reread afterwards as a check that you can reproduce the numbers.

The six numbers worth being able to reproduce cold

Number	Chapter	Why that one
2.8×10^6 proteins, 9×10^8 residues, 1 molecule = 1.66 nM	B1	The denominators of every per-cell calculation
71 kJ/mol for a 10^{12} rate enhancement; 5.93 kJ/mol per decade	B2	The scale of catalysis, and why barriers are small numbers exponentiated
$\Delta p = -177$ mV → 17.1 kJ per H^+; minimum 2.75 H^+/ATP	B4	Thermodynamics setting a floor on stoichiometry
2.39 g glucose per g of cells; 50% of carbon to CO_2	B7	The baseline against which every product yield is judged
1.1% of proteome from a strong gene; 2% growth cost; 20% → 36%	B8, B9	The price of expression
Variance/mean 649 against 1; 5×10^8 mutants per generation	B10	Why every large culture already contains the variant that will take it over

On the numbers in this part

Everything without a † was computed from the standard genetic code, from the recalled constants listed in the table before this note, or from numbers already established in parts 2 to 13. The recalled constants are the weak point, and the ones that carry the most weight are the ribosome parameters (7,500 residues per ribosome, 18 aa/s, 55% active), which set κ_t; $\varphi_{max} = 0.55$, which sets the burden slope; the P/O assumptions, which set the ATP yield; and the 48% carbon content, which sets the 50% carbon-to-biomass figure. If one of those is wrong by 20%, the chapter's conclusion survives and its third digit does not.

I also corrected two statements from earlier parts as I went. Part 1 chapter 6 said succinyl-CoA feeds haem synthesis, which is the route in yeast and animals but not in *E. coli*, which uses glutamate through the C5 pathway (chapter B6). And part 14 uses 7 NADPH + 7 NADH for an *E. coli* palmitate where a shorthand elsewhere in the module says 14 NADPH — the latter is right for yeast (chapter B7).